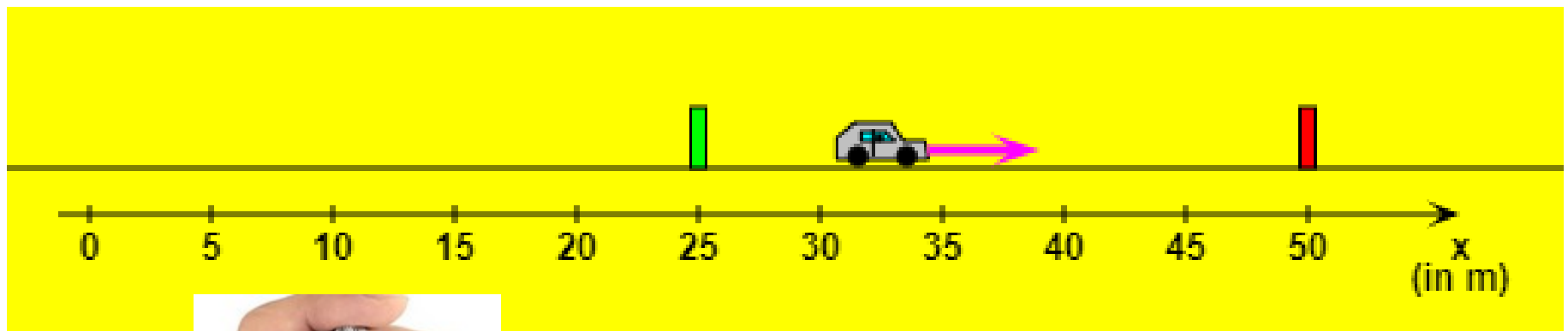
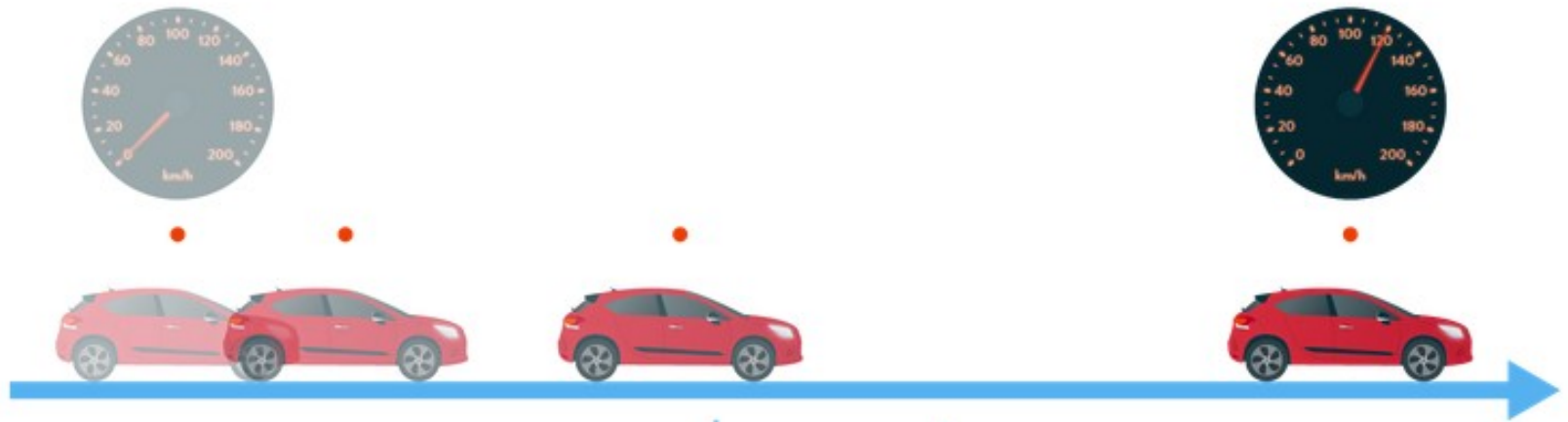
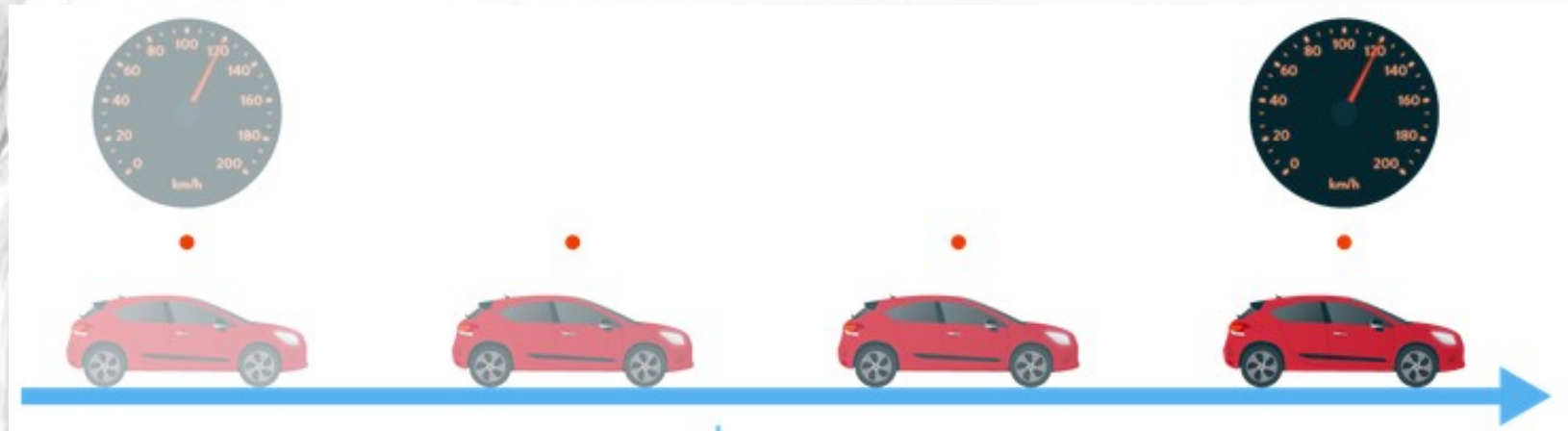


Unit2

Kinematics in One Dimension



discuss



Kinematics deals with the concepts that are needed to describe motion.

Dynamics deals with the effect that forces have on motion.

Together, kinematics and dynamics form the branch of physics known as ***Mechanics***.

Kinematics is the portion of mechanics that describes motion without any reference to ?

- a) forces
- b) accelerations
- c) velocities
- d) displacements
- e) time

Distance - Average speed

Distance (d) measures “how far”. Unit is m.

time (t) is the time elapsed. Unit is second.

Average speed (s) is the distance traveled divided by the time required to cover the distance. Unit is m/s.

$$\text{Average speed} = \frac{\text{total distance}}{\text{elapsed time}}$$

SI units for speed: **meters per second** (m/s)

Example:

A NASCAR car travels at an average speed of 190 miles/hour.

A) How long would a 500 mile race take in hours?

B) Convert 190mph to m/s

(remember $1\text{m/s} = 2.2\text{ mph}$)

C) in 5 hours how many miles were covered ?

Convert to km



Equation sheet (tests)

equations

$s. = d/t$ average speed

$\Delta x = X_f - X_i$ displacement

$V_{av} = \Delta x/t$ average velocity

$\Delta V = V_f - V_i$ change in velocity

$a. = \Delta V/t$ acceleration

Kinematics equations 1D:

1) position x (location on number line)
 $x = x_i + V_i t + 0.5 a t^2$

Or displacement ($x - x_i$ how far)
 $\Delta x = V_i t + 0.5 a t^2$

2) final velocity (instantaneous) $V = V_i + at$

3) acceleration $a = (V - V_i)/t$

4) $V^2 = V_i^2 + 2a(x - x_i)$ or $V^2 = V_i^2 + 2a(\Delta x)$

5) Average velocity $= \Delta x/t$ or $(V + V_i)/2$

Physical quantities

t. time elapsed (seconds) – same as Δt

s. average speed (m/s)

d. distance (m) – always positive – how far

X position (m) on number line. Can be < 0

X_f or X final position (m) **X_i** initial position (m)

ΔX (m) displacement = change in position in 1D.
Can be < 0 if moving @ left or up

V_{av} (m/s) average velocity

V (m/s) velocity . Can be < 0 if moving @ left or up

V_i initial velocity **V_f or V** final velocity

a. acceleration (m/s/s) Can be negative if push/pull @ left

$$\text{Average speed} = \frac{\text{Distance}}{\text{Elapsed time}}$$

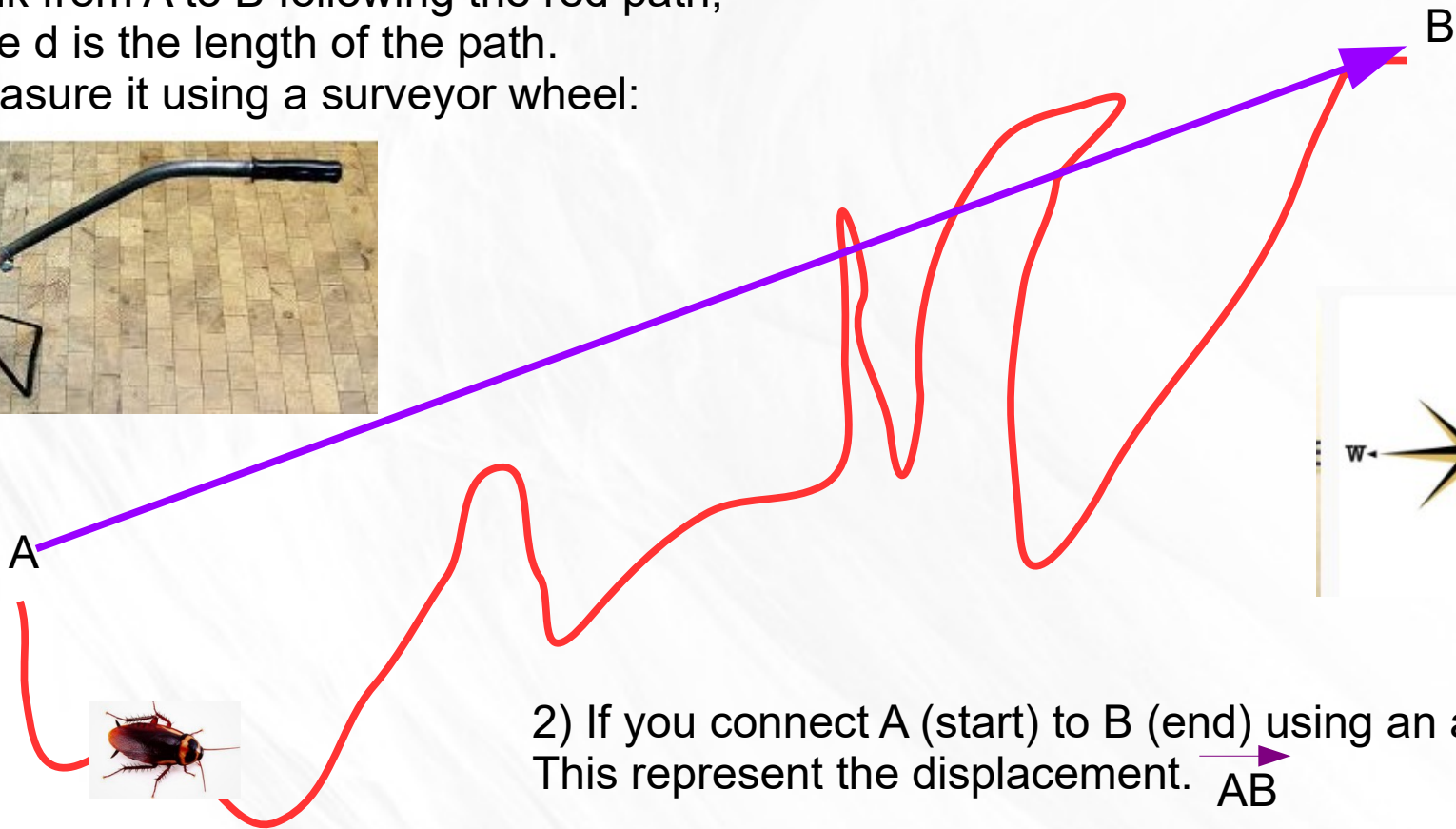
Examples

- 1) How far does a jogger run in 1.5 hours (convert 1st in s) if his average speed is 2.22 m/s? Convert to km
Convert to miles (1 mile = 1,609m)
- 2) If it takes you 45 seconds to run 340m, what is your average speed?
- 3) How long does it take a skier to travel 650m, going 18.5m/s ?
- 4) at 9:00AM you start on a walk. At 10:30AM you are 4.6 miles from Your starting point. what is your average speed in mph ?
Convert to m/s (Sorry units used in Physics are meters, Seconds, kilograms, joule ..) 1 mile = 1609m
- 5) What is the average speed of a runner who finishes the 1,500 meters race in 3 minutes 30 seconds ? In m/s

Your speedometer does not indicate averages; it tells you how fast you are going now. This is your instantaneous speed. If the instantaneous does not change, you are going at a constant speed. The constant value of the speed will be equal to your average speed for the whole trip.

In Physics displacement and distance are not the same physical quantities.

- 1) If you walk from A to B following the red path,
The distance d is the length of the path.
You can measure it using a surveyor wheel:



- 2) If you connect A (start) to B (end) using an arrow
This represent the displacement. $\overrightarrow{AB}$

In Physics/Math
displacement AB is called a vector

A displacement has a length (length of arrow)
and a direction (like 30NW) .

BUT do not worry (for now) in 1D displacement is easy to understand.
Displacement tells if the motion is @ left /@ right or @ up/@down

Coordinate System

1. Origin

2. Axis

positive coordinate direction

3. Unit Vector



Kinematics:

Position as a function of time $x(t)$

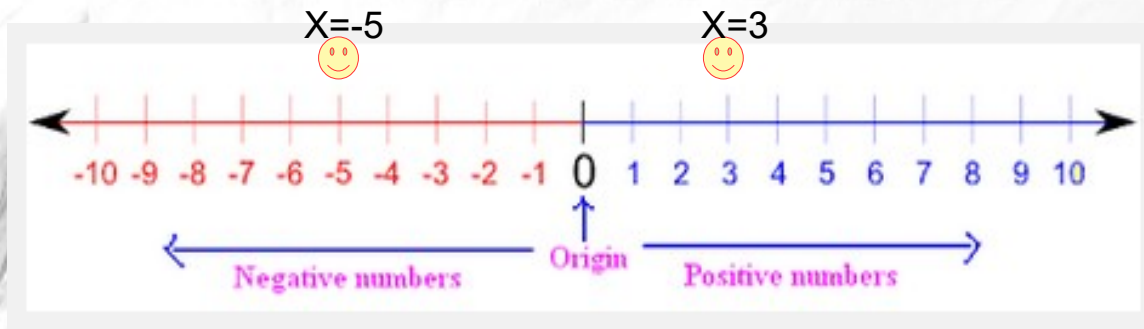
Velocity as a function of time $V(t)$

Acceleration as a function of time $a(t)$

10

Note: mention the lab1. The app with initial position $X_0=10\text{m}$ and Velocity = 5m/s @ right. The equation was $X(t)=5t + 10$

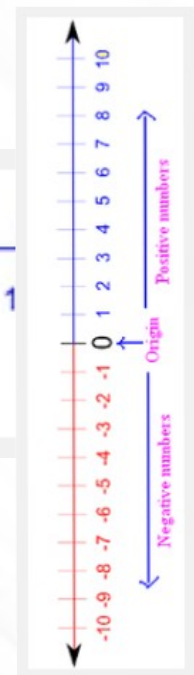
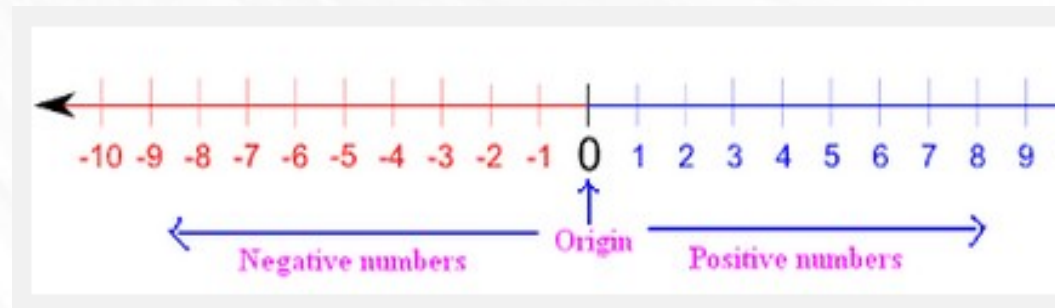
Position x: location on number line



Finding the displacement
In 1D is easy =
Change in position on number line

displacement Δx : change in position in 1D
Connect initial position to final position

$$\Delta x = x_2 - x_1 = -5 - 3 = -8 \text{ units}$$



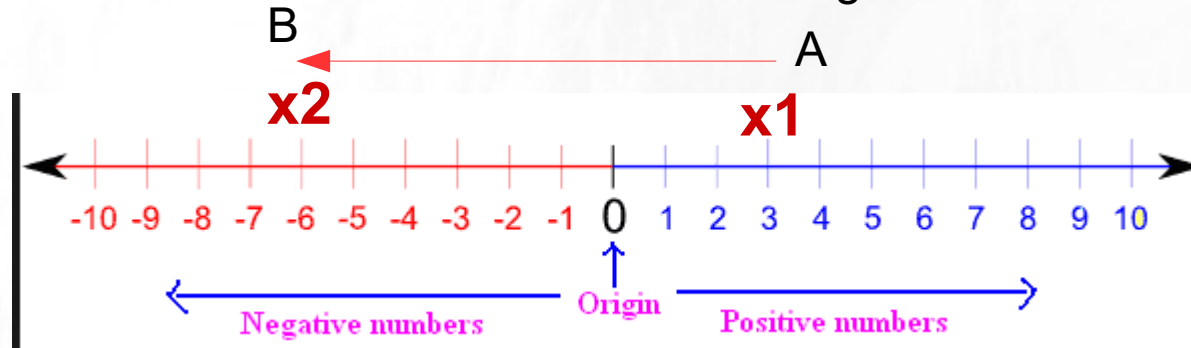
11

If $X_1 = -5$ and displacement = 7 what is the final position X_2 ?

Same principle can apply to the vertical scale.
If $Y_1 = -3$ and the $Y_2 = 2$ what is the displacement

If the Y_1 is -10 and the displacement is 5 what is the final position Y_2 ?

We work in one dimension for now along the number line.



In one dimension the displacement is just the change of position on the number line.

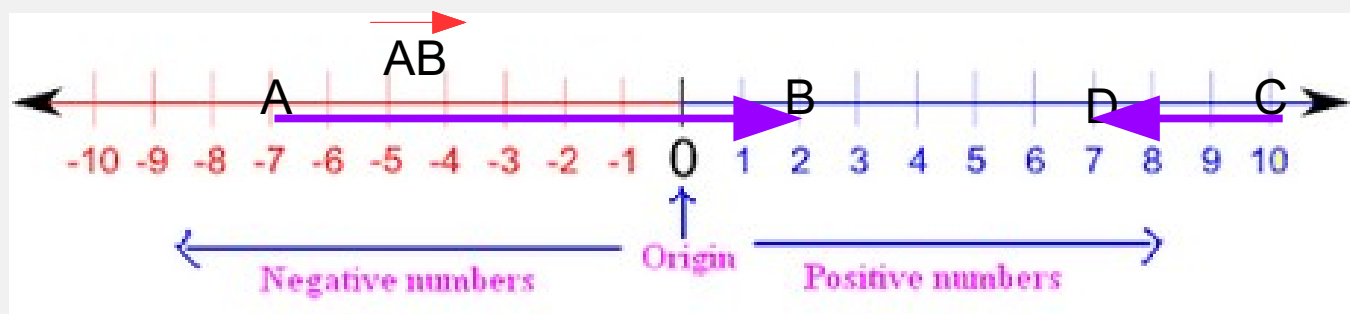
Displacement = final position – initial position

Displacement is noted Δx

$\Delta x = x_2 - x_1 = x_{\text{final}} - x_{\text{initial}}$

Here the displacement $\overrightarrow{AB}$ is $x_2 - x_1 = -6 - 3 = -9$ units.
Motion @ left $\Rightarrow$ the displacement is negative
Motion @ right $\Rightarrow$ the displacement is positive

The distance is always positive. Its called a **scalar**. Here the distance is 9 units.¹²
Displacement in physics is a physical quantity called **vector**.
It has a magnitude, a direction and a unit.



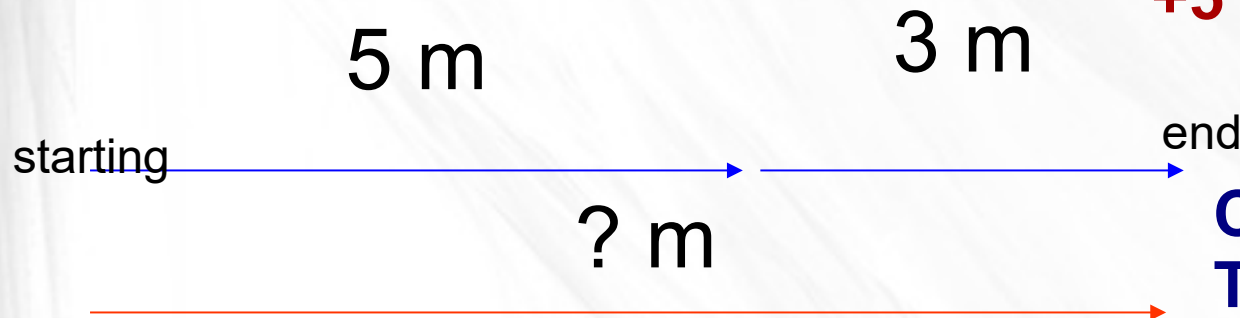
$$AB = \Delta x = ?$$

$$CD = \Delta x = ?$$

If you move from A to B then B to A :

- what is the displacement ? (connect starting point to ending point or $X_{\text{final}} - X_{\text{initial}}$)
- what is the distance ?

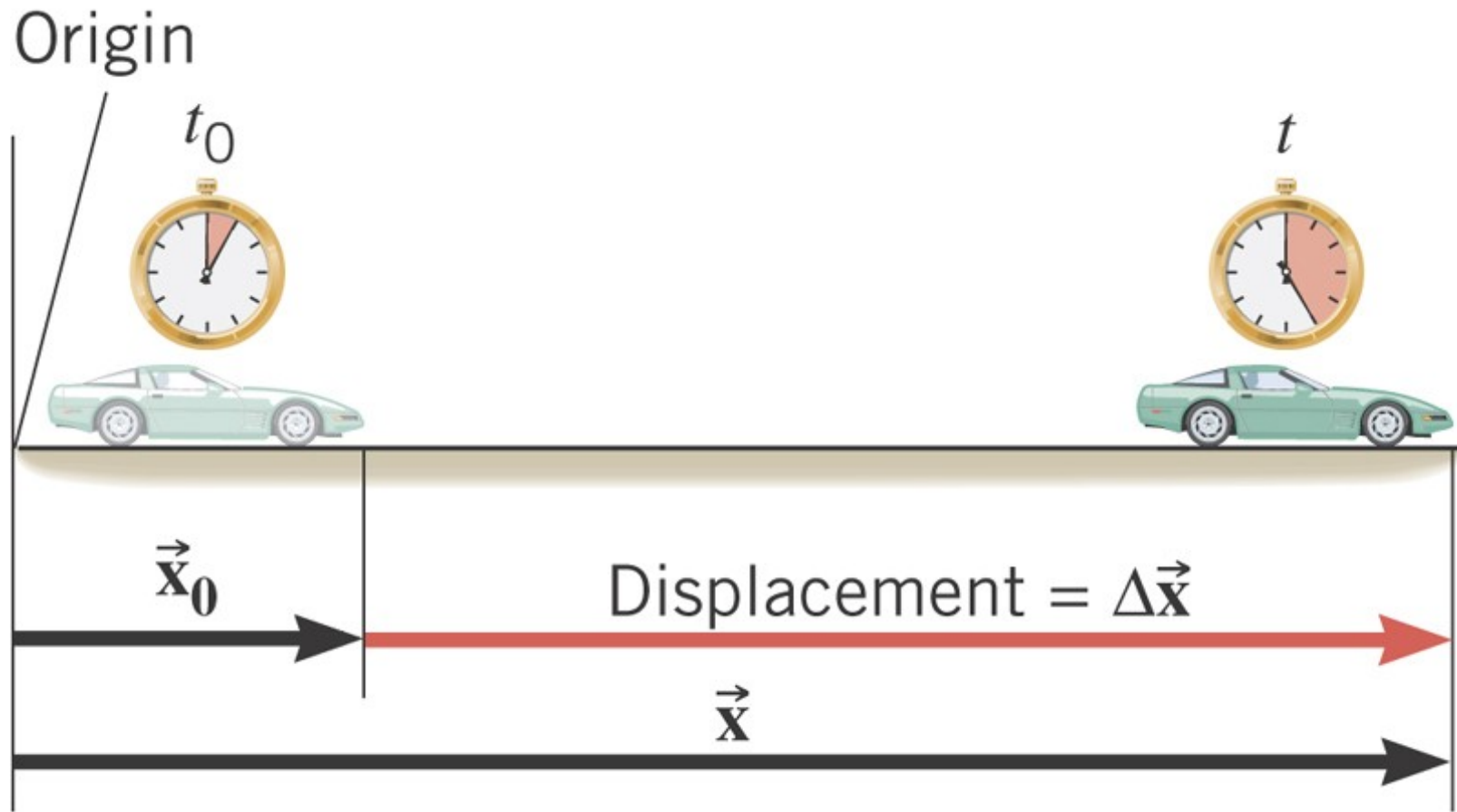
You can add displacements
 $+5 + 3 = +8$



**Connect the starting point
To the final point**

What is the sum of the displacements ?
 What is the notation ?

2.1 Displacement



$\vec{x}_o$ = initial position

$\vec{x}$ = final position

$$\Delta\vec{x} = \vec{x} - \vec{x}_o = \text{displacement}$$

A physical quantity that has a direction and a magnitude like a displacement is called a vector. It is represented by an arrow.

$$\begin{array}{c} \textcolor{red}{5} \\ \longrightarrow \end{array} + \begin{array}{c} \textcolor{red}{5} \\ \longrightarrow \end{array} = \begin{array}{c} \textcolor{red}{10} \\ \longrightarrow \end{array}$$

It is easy to add vectors in 1D. Complete the page.
Draw the vector sum and write its notation.

$$\begin{array}{c} \textcolor{red}{5} \\ \longrightarrow \end{array} + \begin{array}{c} \textcolor{red}{-5} \\ \longleftarrow \end{array} =$$

$$\begin{array}{c} \textcolor{red}{5} \\ \longrightarrow \end{array} + \begin{array}{c} \textcolor{red}{10} \\ \longrightarrow \end{array} =$$

You walk 5 yards @ right and another 5 yards @ right .
The net sum is **10 yards @ right**

$$\begin{array}{c} \textcolor{red}{5} \\ \longrightarrow \end{array} + \begin{array}{c} \textcolor{red}{-10} \\ \longleftarrow \end{array} =$$

10 yards @ left is simply -10 yards.

Since there are only 2 directions. Minus is for left.

$$\begin{array}{c} \textcolor{red}{5} \\ \longrightarrow \end{array} + \begin{array}{c} \textcolor{red}{-15} \\ \longleftarrow \end{array} =$$

Up is positive
Down is negative.

$$\begin{array}{c} \textcolor{red}{10} \\ \uparrow \end{array} + \begin{array}{c} \textcolor{red}{-5} \\ \downarrow \end{array} =$$

$$\begin{array}{c} \uparrow \\ 500 \end{array} + \begin{array}{c} \downarrow \\ 2000 \end{array}$$

$$\begin{array}{c} \uparrow \\ 500 \end{array} + \begin{array}{c} \downarrow \\ 1500 \end{array}$$

$$\begin{array}{c} \downarrow \\ 4000 \end{array} + \begin{array}{c} \uparrow \\ 500 \end{array}$$



Women Who Lift Weights

In Physics Velocity and speed are not the same physical quantity:

average velocity = displacement / time : $V = \Delta x / \text{time}$

If the object is moving @right, velocity is positive

If the object is moving @left, velocity is negative

So to find the velocity : find the displacement (see previous slides) and divide by the time !

velocity is called a vector – notation is $\vec{V}$

It has a magnitude, a direction (which way: left or right if 1D motion), a unit (m/s)

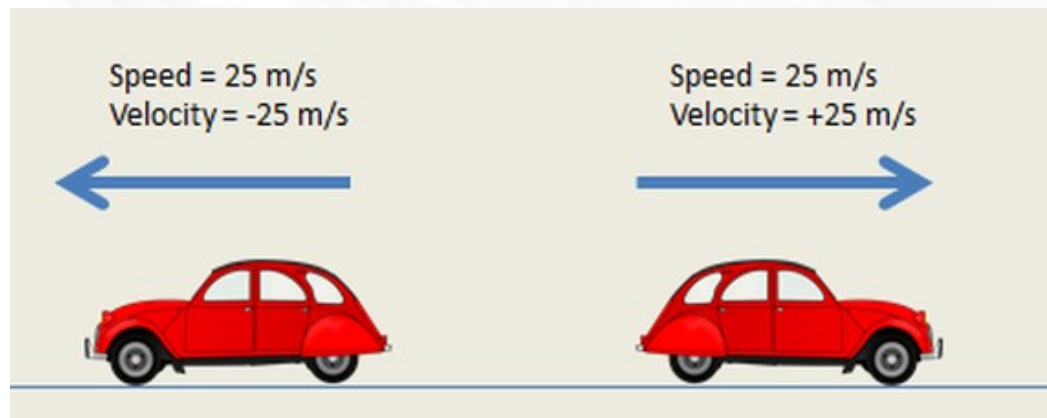
(other examples of vectors: force, acceleration, displacement)

average speed = distance / time = d/t (see previous slides)

speed is not a vector (it is called a scalar) – it has a magnitude, a unit (m/s)

(other examples of scalars: time, energy, distance, temperature, volume ...

any quantity without a direction)



2.2 Speed and Velocity

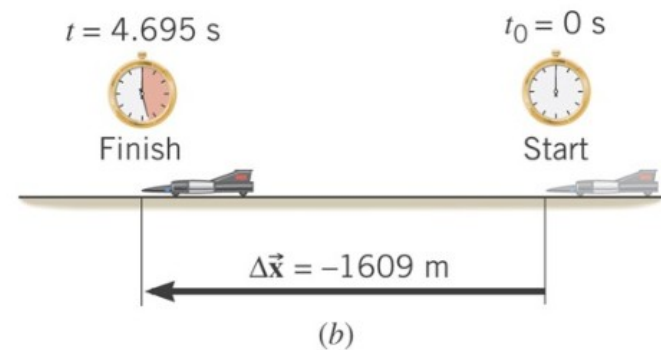
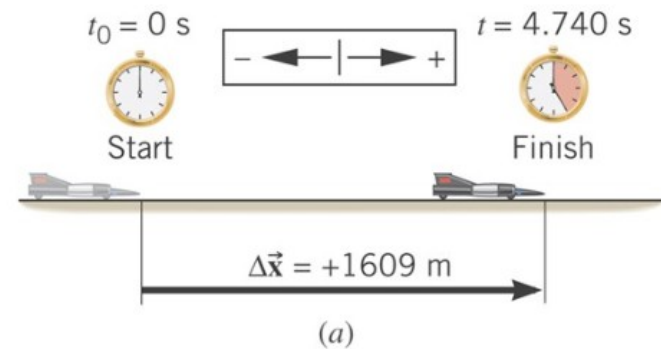
Average velocity is the displacement divided by the elapsed time.

$$\text{Average velocity} = \frac{\text{Displacement}}{\text{Elapsed time}}$$

$$\vec{v} = \frac{\vec{x} - \vec{x}_o}{t - t_o} = \frac{\Delta \vec{x}}{\Delta t}$$

Example 2 The World's Fastest Jet-Engine Car

Andy Green in the car *ThrustSSC* set a world record of 341.1 m/s in 1997. To establish such a record, the driver makes two runs through the course, one in each direction, to nullify wind effects. From the data, determine the average velocity for each run.



Displacement = how far you are from your starting point and @ left or @ right.

Distance = total distance covered

1) A race car covers 1700m @ right in 5s. What is the average velocity?

The same race drive in the opposite direction with the wind opposing

Its motion. It covers 1700m @ left in 5.4 s. What is the average velocity ?

2) In NYC you walk 10 blocks @ south (or -10) in 30minutes. What is your average velocity in blocks/hour

In class

3) you drive 100 miles @ east (so +10) in 2 hours.

What is your average velocity in mph and m/s

4) If you walk 5 blocks @ north and you realize this is the wrong way.

You then walk 10 blocks @ south. The all trip lasted 30 minutes.

In class

What is the average velocity (from start to end) blocks/hour

What is the average speed. (not the same)

5) same question for 10 blocks @ west (-10) and 20 blocks @ east.

The trip lasts 45 minutes

6) same question for 100 miles @ North and 50 miles @ south in 2 hours.

In mph in m/s

7) An insect flies 5 inches @ up 3 inches down then 15 inches @ up in 2 s.

What is its average velocity ? In in/s Average speed?

Δx = displacement

V = velocity

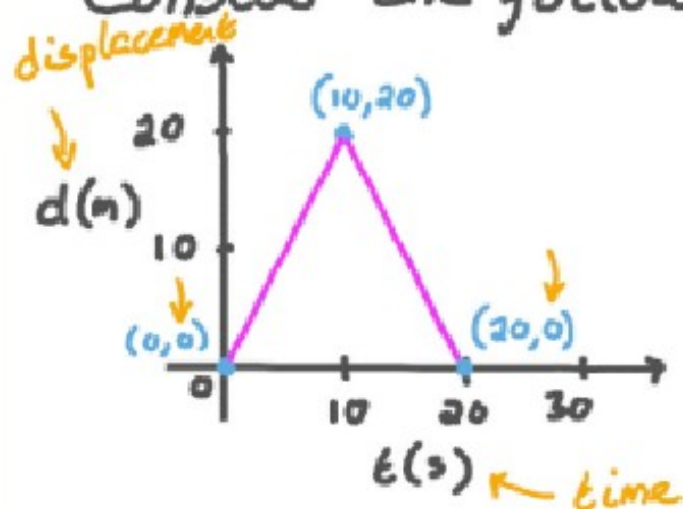
$V = \Delta x / t$

d = distance

S = speed

$S = d / t$

Consider the following graph.



$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

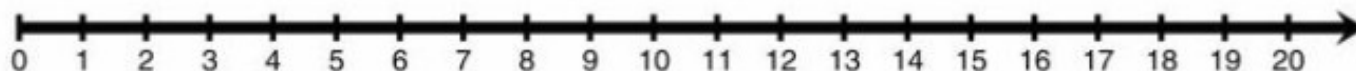
$$= \boxed{}$$

$$\text{Average velocity} = \frac{\text{Net displacement}}{\text{Total Time}}$$

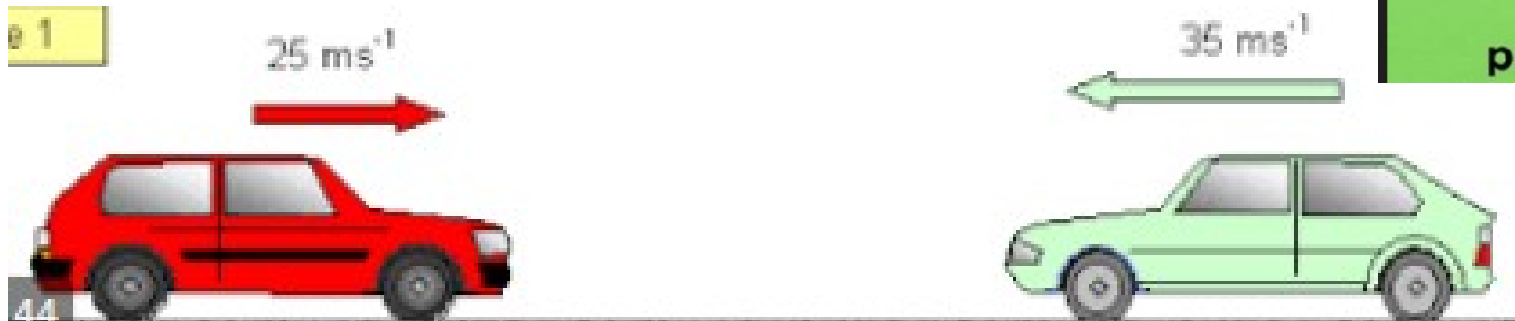
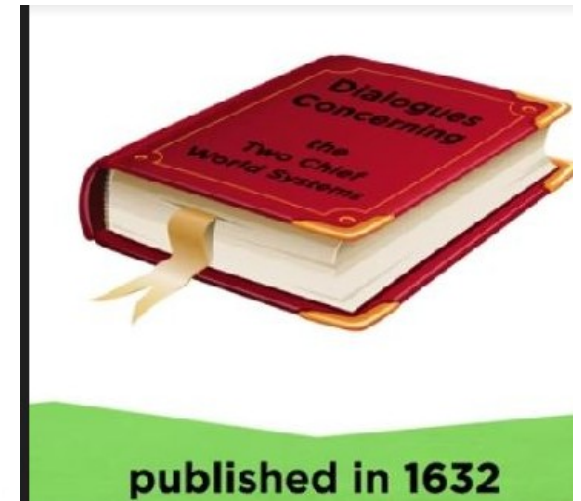
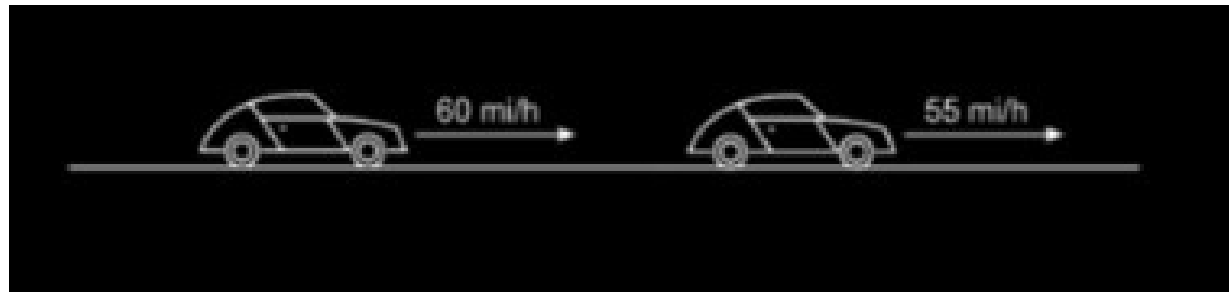
$$= \boxed{}$$

Complete the following: The average speed is $\boxed{}$ m/s.

Complete the following: According to the graph, the average velocity is $\boxed{}$ m/s.



Relative velocity



$$V_{car/u} = V_{car/gnd} - V_{u/gnd}$$

If the car in front of you is moving at the same velocity as you/
Relative to you, its speed is / If the car in front of you slows down
Relative to you, the car is moving toward? Away? Same distance?
Example: sky diver opening a parachute filmed by a sky diver without a chute.

VELOCITY VECTORS



Swimming by Skills NT

Counter-current Pools: Alternative to...

Relative velocity

A barge leaves the dock, heading away at 3 mi/h. A man on the barge, facing away from the dock, walks at 2 mi/h relative to the barge. What is the man's speed relative to the dock?

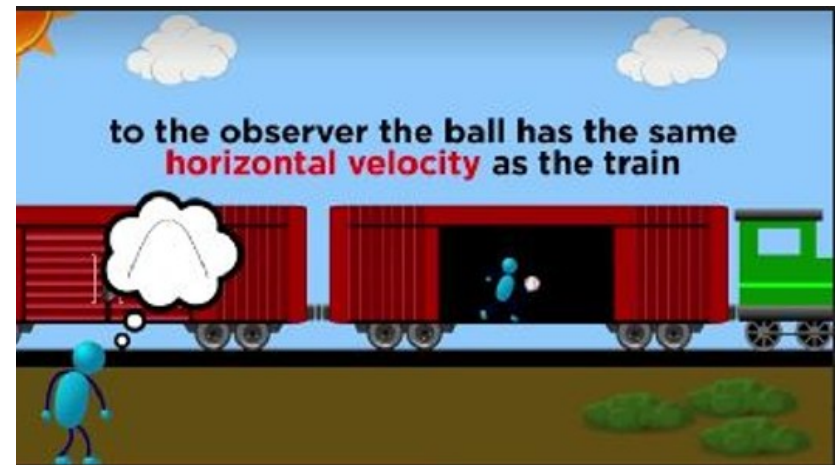


$$V_{\text{man/dock}} = V_{\text{man/boat}} + V_{\text{dock/boat}}$$

A barge leaves the dock, heading away at 3 mi/h. A man on the barge, facing toward from the dock, walks at 2 mi/h relative to the barge. What is the man's speed relative to the dock?



$$V_{\text{man/dock}} = V_{\text{man/boat}} + V_{\text{boat/dock}}$$



Discuss relative speed if both train and person has the same speed but opposite See video



Is the speed constant ?
Is he speeding up or slowing down?

We can still find the average acceleration
Average velocity (displacement/time)
You see the average speed on graph.

ACCELERATION = $\frac{\text{CHANGE IN SPEED}}{\text{TIME}}$

$$\frac{31.5 \text{ ft/s}}{1.2 \text{ sec}}$$

ACCELERATION = 26.25 ft/s^2

I'm accelerating because I'm **speeding up**.

I'm accelerating because I'm **slowing down**.

I'm accelerating because I'm **changing directions**.

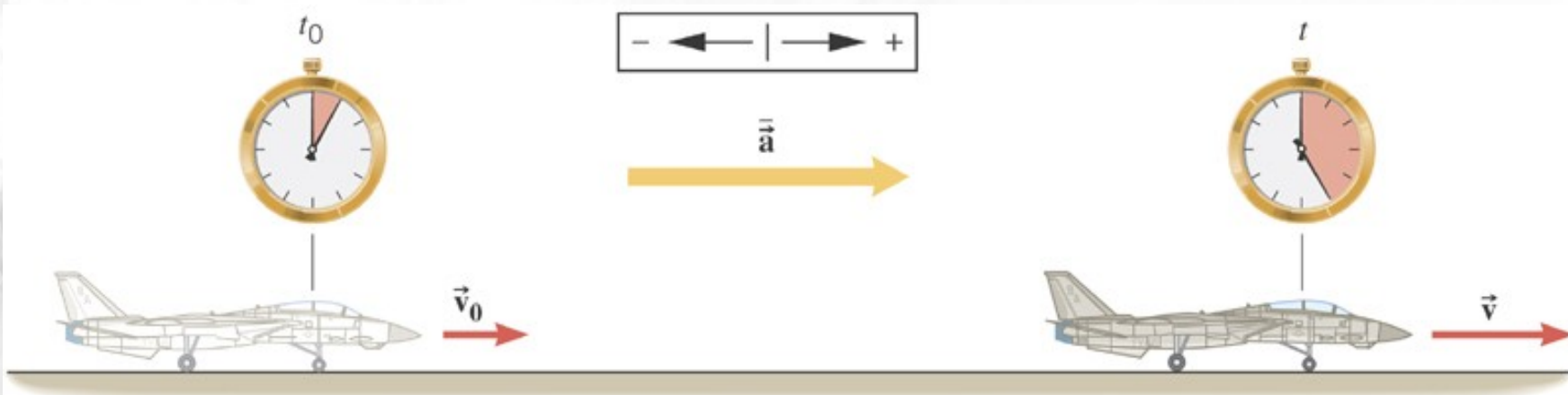
Which way is:
Displacement
Velocity
Acceleration?

I'm accelerating because I'm **speeding up**.

I'm accelerating because I'm **slowing down**.

I'm accelerating because I'm **changing directions**.

First step in kinematics: the number line or Coordinate system.



DEFINITION OF AVERAGE ACCELERATION

(how fast an object is getting faster)

a = change of velocity / time elapsed = (Vfinal – Vinitial) / time
acceleration is the change of velocity per unit time

Unit is (m/s)/s or m/s/s or m/s²

Remember: acceleration has the same direction as the force applied to the object!
Because acceleration and force are proportional,

Example: acceleration due to gravity is about 22mph/s

Which means that every second a falling object goes 22mph faster every second.

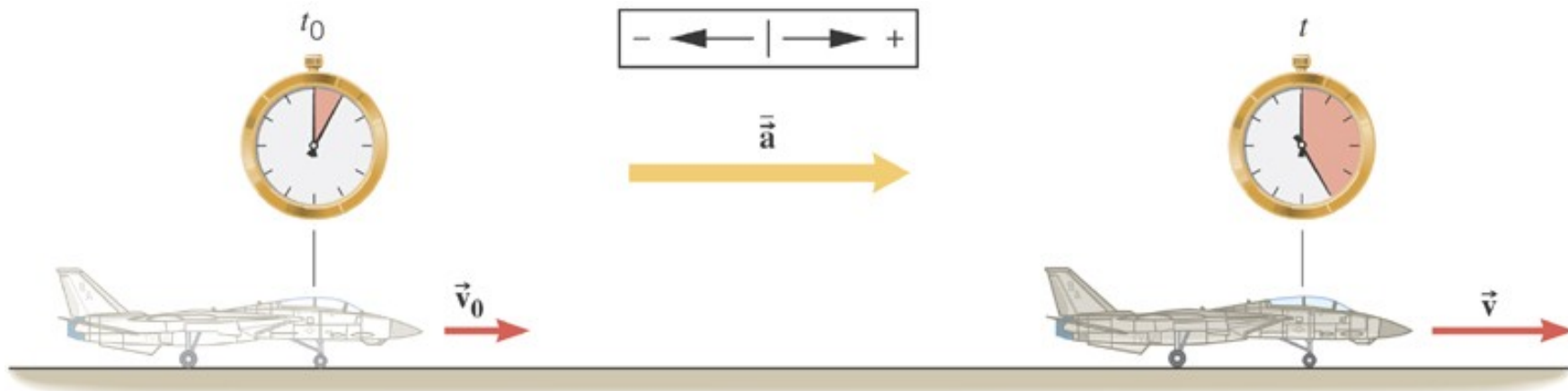
A falling object (dropped from a tall tower) reaches a speed of 88mph after ___ seconds.

The acceleration is @ down because the _____ of the object is pulling it down.

22mph per second is _____ m/s per second.

After 3 seconds of fall, the speed reached is _____ m/s

2.3 Acceleration



DEFINITION OF AVERAGE ACCELERATION

$$\vec{a} = \frac{\vec{v} - \vec{v}_o}{t - t_o} = \frac{\Delta \vec{v}}{\Delta t}$$

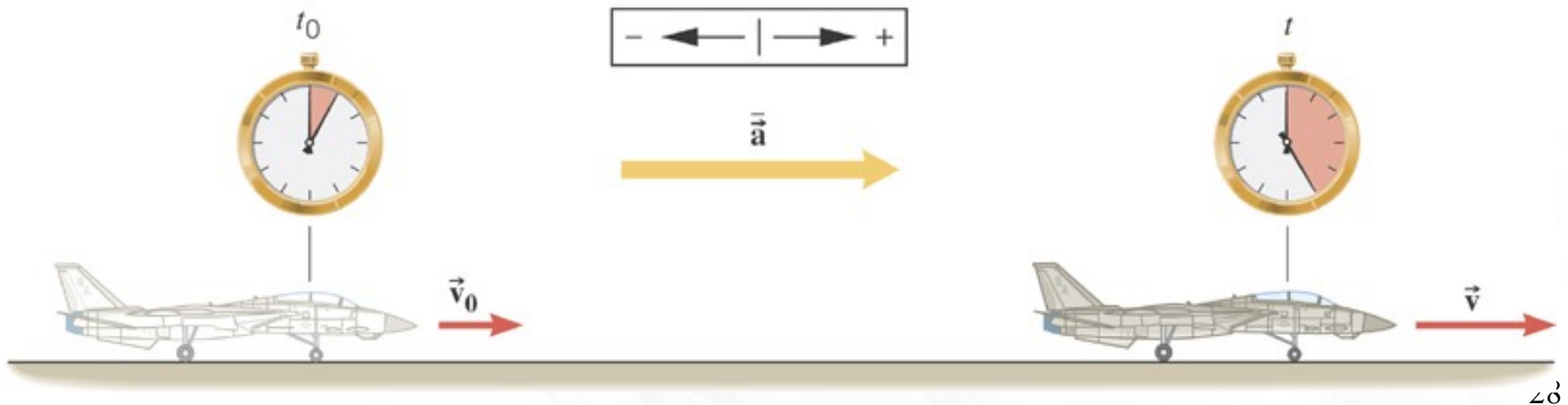
ACCELERATION is a vector.

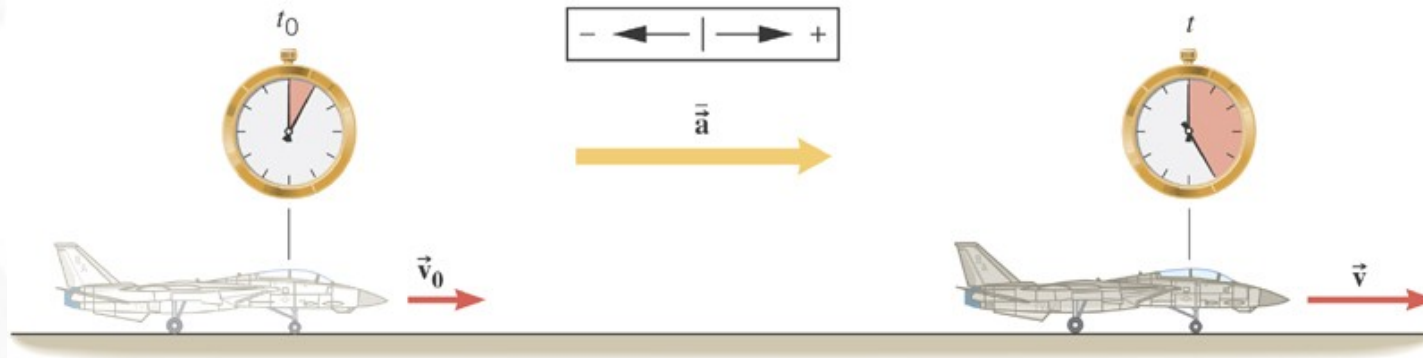
It has a magnitude (change in velocity per unit time) and a direction

If there is a thrust (or push/pull) @ right, acceleration is positive

If there is a thrust @ left, acceleration is negative.

The notion of *acceleration* emerges when a change in velocity is combined with the time during which the change occurs.





Example Acceleration and Increasing Velocity

Determine the average acceleration of the plane in km/h/s

$V_{\text{initial}} = 0$

$V_{\text{final}} = 250 \text{ km/h}$

time elapsed = 30s

A) Acceleration in km/h/s (see next slide before going to B)

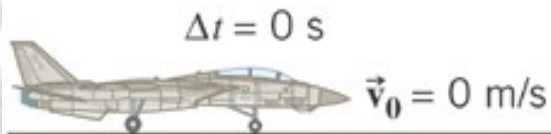
B) How fast it is going after 5 seconds (find speed in km/h)

C) for the same acceleration what is the speed after 5s if the initial velocity is 50km/h ?

D) convert the speed found in B to m/s (1km=1000m and 1 hour=3600s)

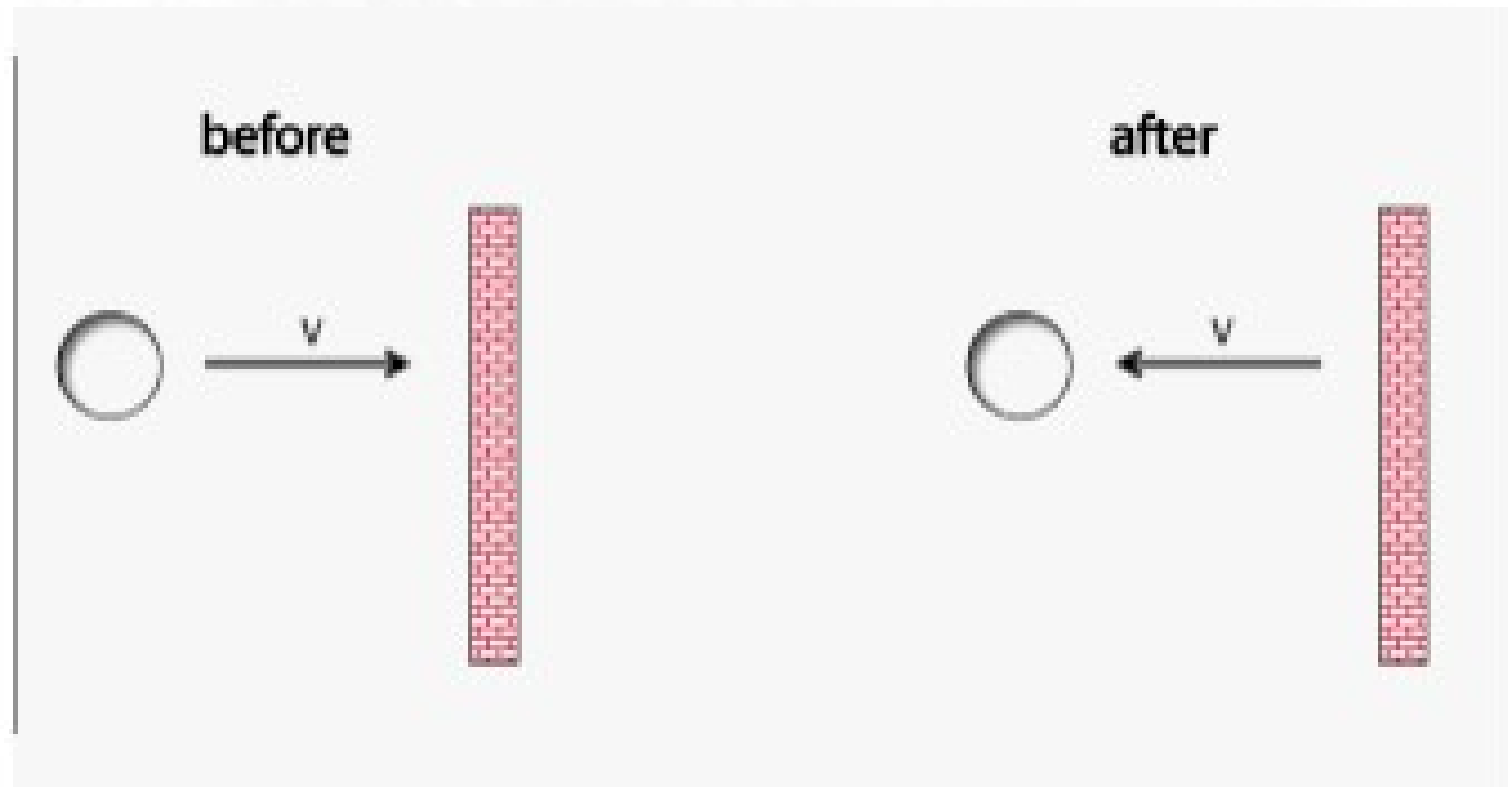
$$\vec{a} = + 8 \frac{\text{km/h}}{\text{s}}$$

Can you find the rate of Increase?



Acceleration is a vector:
8 is the magnitude ; km/h/s is the unit ; and + is the direction

Bouncing ball. $v=5\text{m/s}$ (magnitude) what is the change in velocity? Acceleration?
Which way the force from the wall is acting?



http://www.walter-fendt.de/html5/phen/acceleration_en.htm

$a=-1$ show anatomy

Acceleration and Decreasing Velocity

Initial velocity = 28 m/s

Final velocity = 13 m/s

Between $t=9\text{ s}$ and $t=12\text{ s}$

Find the acceleration

1. 5 m/s/s

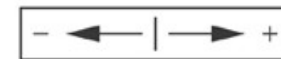
2. 1.7 m/s/s

3. -5 m/s/s

The acceleration has the same direction
than the force (push/pull)

Here the pull is @ left so acceleration is @ ____
but the velocity is @ _____

Velocity is a vector too. It has a _____
And a _____.

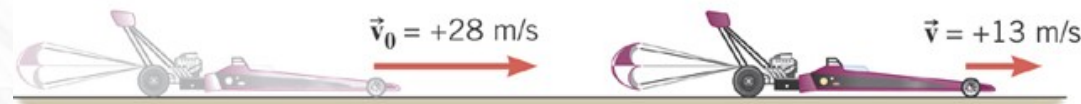
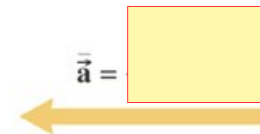


$t_0 = 9.0\text{ s}$



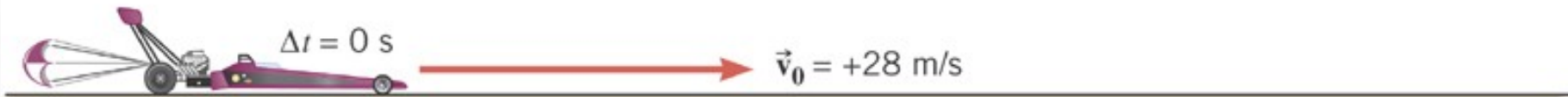
$t = 12\text{ s}$

$\vec{a} =$



(b)

$$\vec{a} = -5.0 \text{ m/s}^2$$

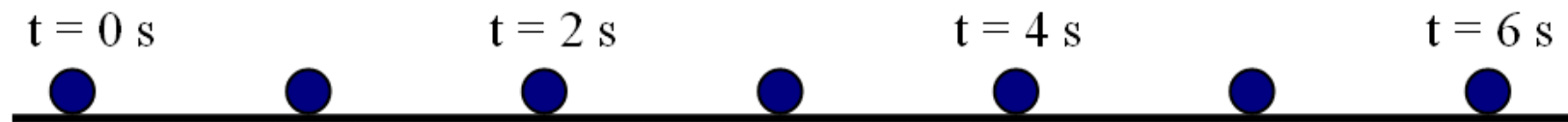


If velocity and acceleration have the same direction => speeding up
If velocity and acceleration have opposite direction => slowing down

Starting from rest, a particle confined to move along a straight line is accelerated at a rate of 2 m/s^2 .

- a) The particle travels 2 m during each second.
- b) The particle travels 2 m only during the first second.
- c) The speed of the particle increases by 2 m/s during each second.
- d) The acceleration of the particle increases by 2 m/s^2 during each second.

The drawing shows the position of a rolling ball at one second intervals. Which one of the following phrases best describes the motion of this ball?



- a) constant position
- b) constant velocity
- c) increasing velocity
- d) constant acceleration
- e) decreasing velocity

In which one of the following situations does the car have an acceleration that is directed due north?

- a) A car travels northward with a constant speed of 24 m/s.
- b) A car is traveling southward as its speed increases from 24 m/s to 33 m/s.
- c) A car is traveling southward as its speed decreases from 24 m/s to 18 m/s.
- d) A car is traveling northward as its speed decreases from 24 m/s to 18 m/s.
- e) A car travels southward with a constant speed of 24 m/s.

A postal truck driver driving due east gently steps on her brake as she approaches an intersection to reduce the speed of the truck. What is the direction of the truck's acceleration, if any?

- a) There is no acceleration in this situation.
- b) due north
- c) due east
- d) due south
- e) due west

A sports car starts from rest. After 10.0 s, the speed of the car is 25.0 m/s. What is the magnitude of the car's acceleration?

- a) 2.50 m/s^2
- b) 5.00 m/s^2
- c) 10.0 m/s^2
- d) 25.0 m/s^2
- e) 250 m/s^2

Which one of the following is not a vector quantity?

- a) acceleration
- b) displacement
- c) instantaneous velocity
- d) average velocity
- e) average speed

Which one of the following equations is the correct expression for average acceleration?

a) $a = \frac{v}{t}$

b) $a = \frac{dv}{dt}$

c) $a = \frac{\Delta x}{\Delta t}$

d) $a = \frac{\Delta v}{\Delta t}$

e) $a = \frac{1}{2} vt^2$

- 1) What is the acceleration of a rocket ship in outer space that takes 5.0s to increase its speed from 1240m/s to 1300m/s ?
(textbook)
- 2) In outer space a rocket ship is traveling at the enormous speed of 2800m/s. What is its acceleration if it increases its
Speed uniformly and is going 2840m/s after 25 s ?
Convert to mph (do it in class)
- 3) A car travels with an average speed of 25m/s (do it in class)
A) What is the speed in km/s (1km =1000m)
B) What is the speed in km/h (1m/s = 3.6 km/h)
- 4) A car travels with an average speed of 58mph. What is the speed in km/h
(1.6km = 1 mile)
- 5) Starting from rest and moving in a straight line, a runner achieves a velocity of 7m/s in a time of 2s. What is the
Average acceleration of the runner ?
- 6) starting from rest, a car accelerates at a rate of 4.2m/s/s for a time of 5 seconds.
What is its velocity at the end of this time ?
- 7) The velocity of a car decreases from 30m/s to 18m/s in a time of 4seconds. What is the average
acceleration if the car
In this process ?

Equations of kinematics: the big 5

Notations:

V (V_{final}) final velocity

V_i (V_{initial}) initial velocity $\rightarrow$ velocity at $t=0$

X_i (X_{initial}) initial position $\rightarrow$ position at $t=0$

X (X_{final}) final position

Δx is the displacement ($x-x_i$) how far you are from "home"

a acceleration

t is the time elapsed

V_i is the initial velocity and is
Sometimes noted V_0

X_i is the initial position (home) and
Is sometimes notes x_0

EQUATIONS OF KINEMATICS

for motion with constant acceleration

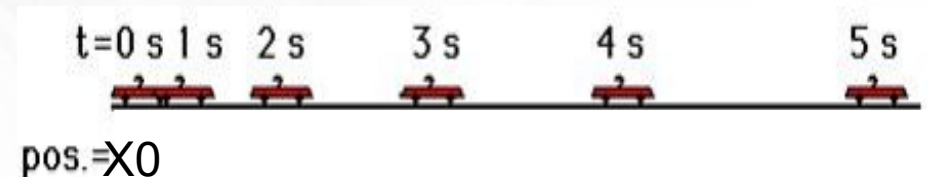
1) $x = x_i + V_i t + 0.5 a t^2$ or $\Delta x = V_i t + 0.5 a t^2$ if $x_i = 0$ then $x = V_i t + 0.5 a t^2$

2) $V = V_i + at$

3) $a = (V - V_i)/t$

4) $V^2 = V_i^2 + 2a(x - x_i)$ or $V^2 = V_i^2 + 2a(\Delta x)$

5) Average velocity $= \Delta x/t$ or $(V + V_i)/2$



EQUATIONS OF KINEMATICS

for motion with constant velocity and acceleration $= 0$

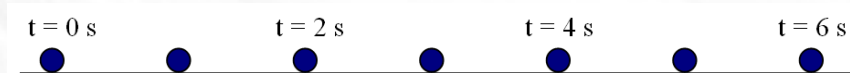
1) $x = x_i + V_i t$ or $\Delta x = V_i t$ if $x_i = 0$ then $x = V_i t$

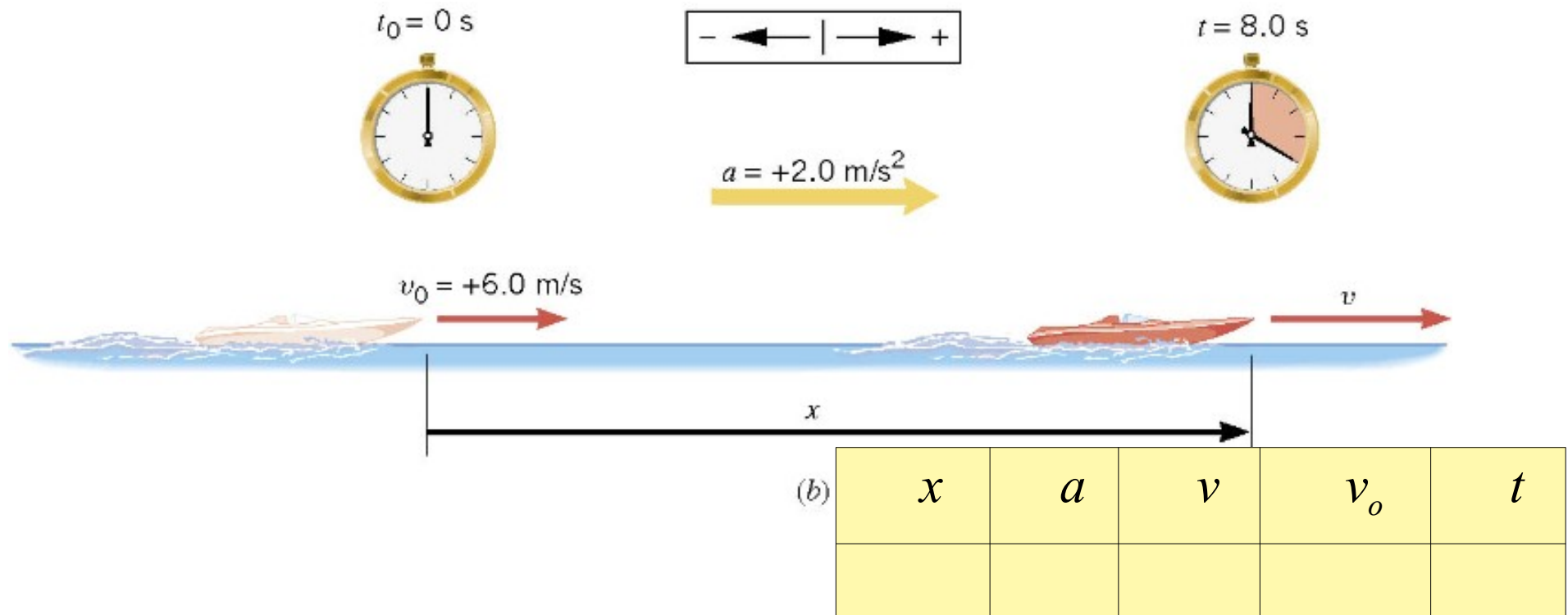
2) $V_i = V_f$ stays the same

3) $a = 0$

4) NA

5) Average velocity $= \Delta x/t$





FIND THE displacement of the boat :

1. 64m

2. 72m

3. 76m

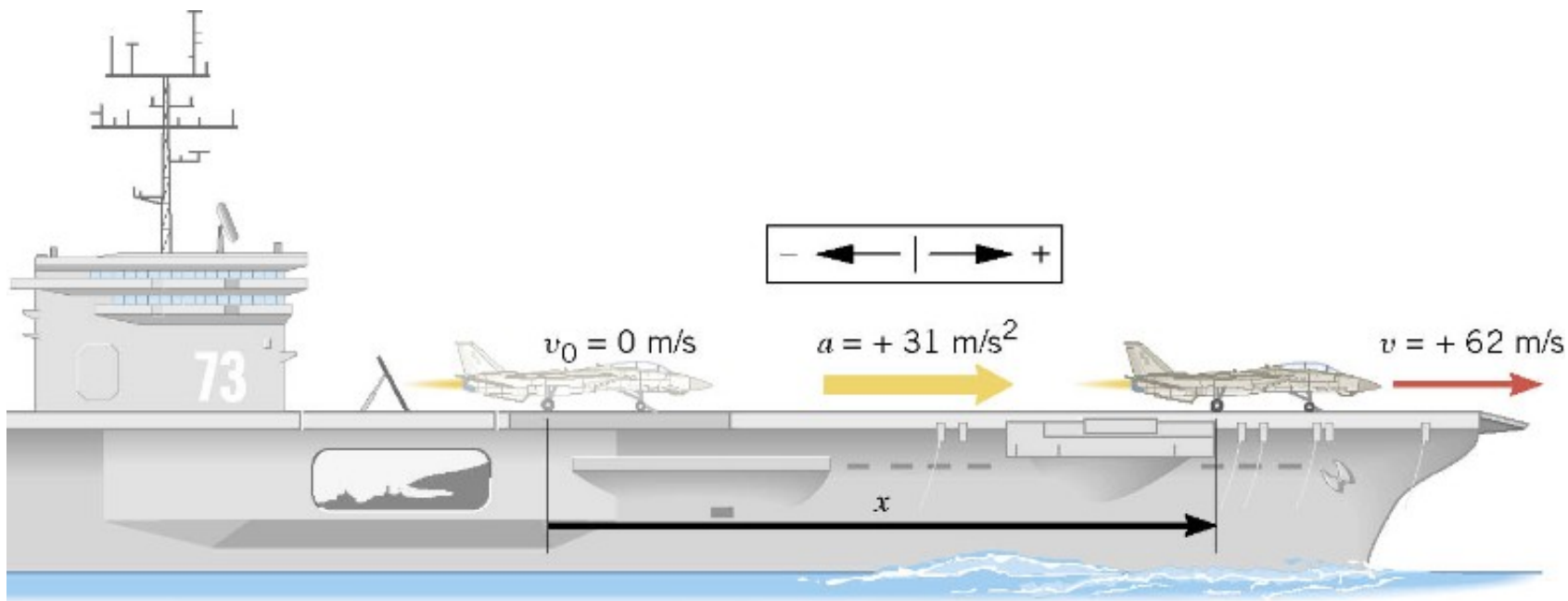
4. 112m

$$[1] \quad x = v_0 t + \frac{1}{2} a t^2$$

$$[2] \quad v = v_0 + a t$$

$$[3] \quad a = \text{constant}$$

$$[4] \quad v^2 = v_0^2 + 2a(x)$$



FIND THE displacement of the airplane :

1. 62m
2. 1m
3. 124m

x	a	v	v_o	t

[1] $x = v_0 t + \frac{1}{2} a t^2$

[2] $v = v_0 + a t$

[3] $a = \text{constant}$

[4] $v^2 = v_0^2 + 2a(x)$

2.4.3. In which one of the following situations is the displacement of the ball directly proportional to the elapsed time?

- a) a ball rolls with constant velocity
- b) a ball at rest is given a constant acceleration
- c) a ball rolling with velocity v_0 is given a constant acceleration
- d) a ball rolling uphill experiences a decreasing acceleration
- e) a ball rolling downhill experiences an increasing acceleration

Example 8 An Accelerating Spacecraft

A spacecraft is traveling with a velocity of +3250 m/s. Suddenly the retrorockets are fired, and the spacecraft begins to slow down with an acceleration whose magnitude is 10.0 m/s². What is the velocity of the spacecraft when the displacement of the craft is +215 km, relative to the point where the retrorockets began firing?

x	a	v	v_o	t

2 possible answers.

1. +/- 6262500 m/s
2. +/- 2502.5 m/s
3. +/- 3855 m/s

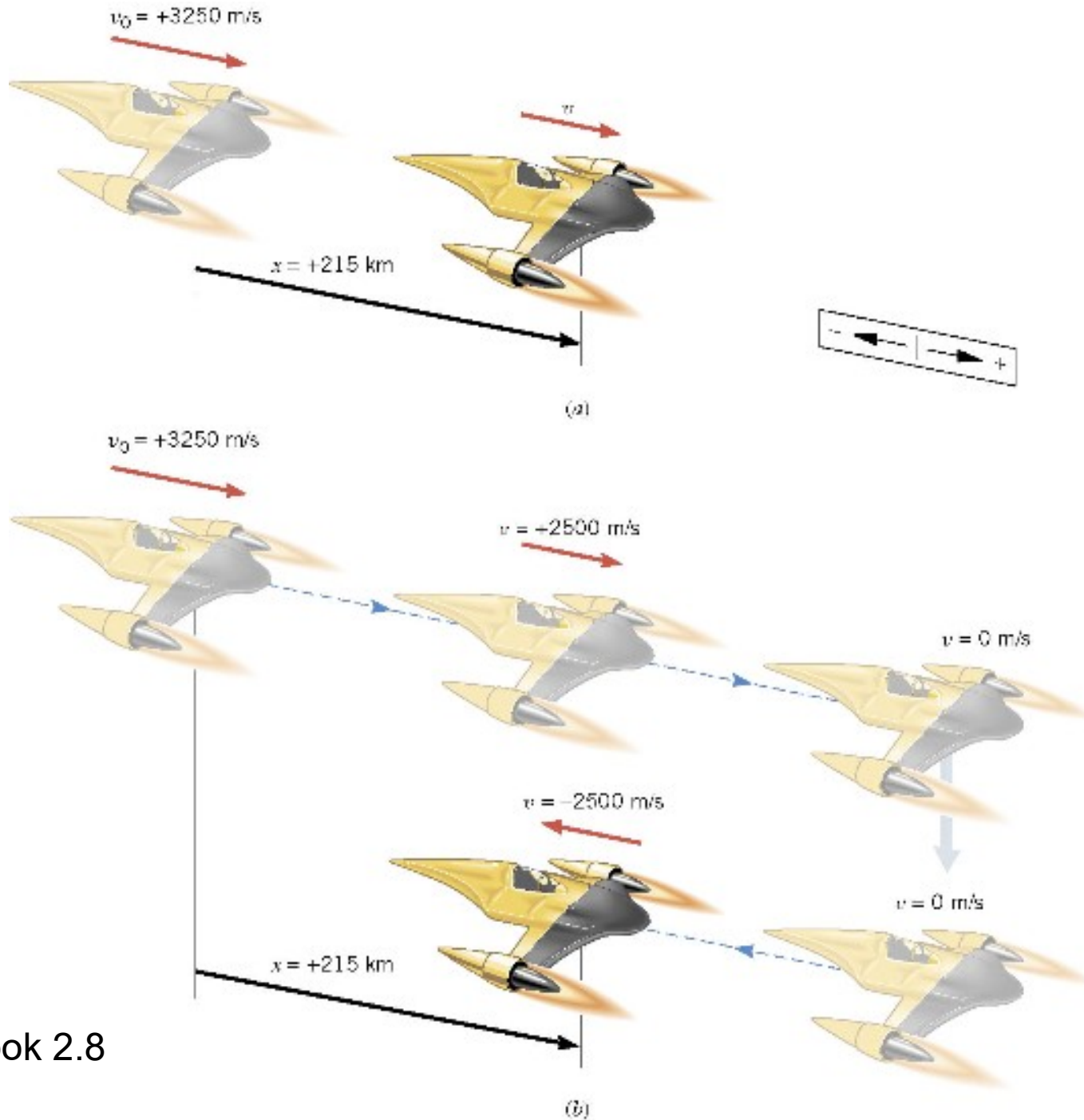
$$[1] \quad x = v_o t + \frac{1}{2} a t^2$$

$$[2] \quad v = v_o + a t$$

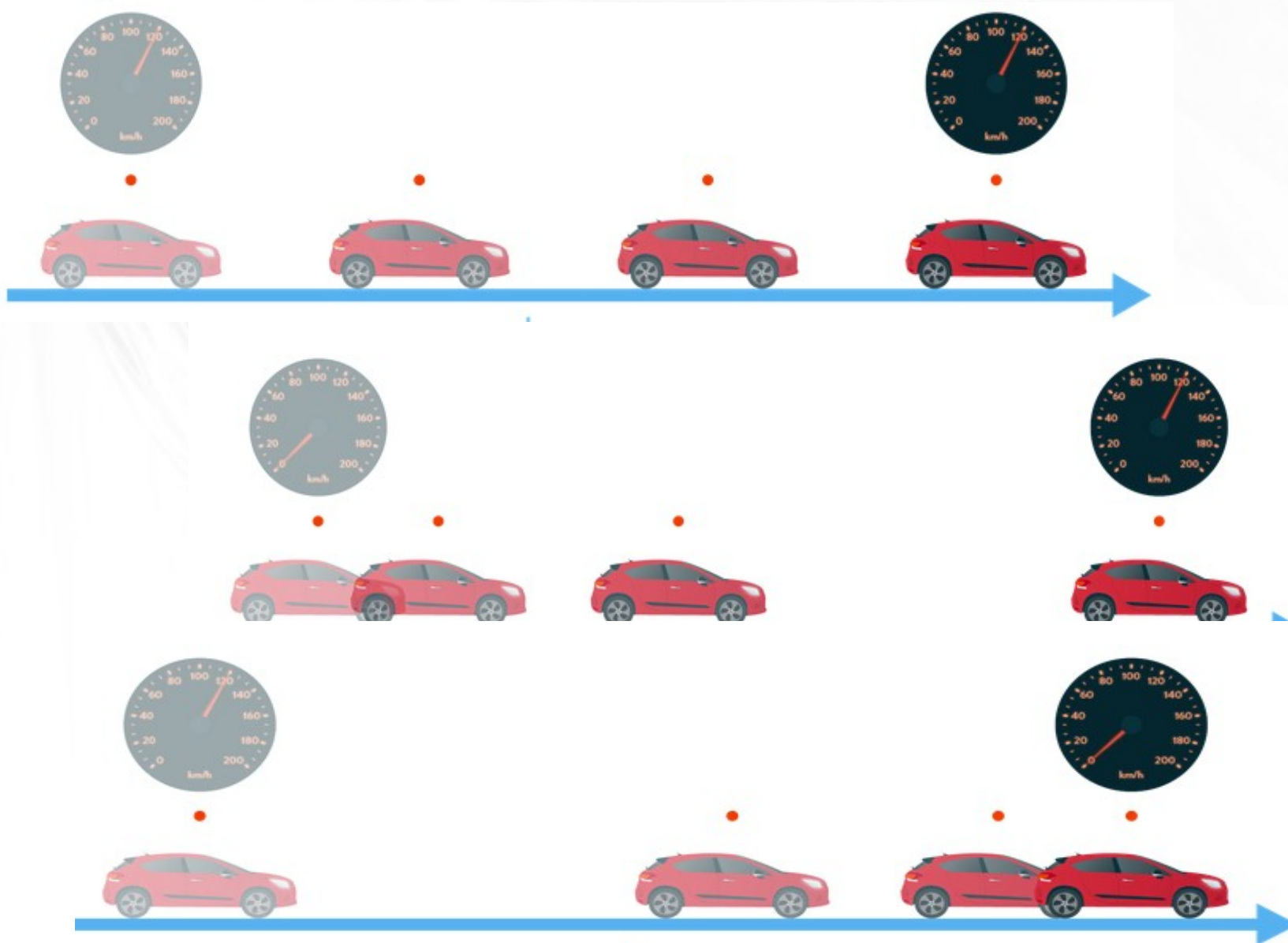
$$[3] \quad a = \text{constant}$$

$$[4] \quad v^2 = v_o^2 + 2a(x)$$

Why 2 possible answers ?



In which case $a < 0$, in which case $a > 0$, in which case $a = 0$
What about velocity.



**Let's use the 2 equations : $\Delta x = v_i t + 0.5 (a) t^2$ and $V = v_i + at$
or $x = x_i + v_i t + 0.5 (a) t^2$**

- 1) A car accelerating with an initial velocity of 12m/s accelerates at a constant rate of 2.5m/s/s for a time of 2s.
 - A) What is its velocity at the end of this time ?
 - B) What displacement does the car cover during this process (the initial position $X_0 = 0$ find X)

- 2) A car moving with an initial velocity of 30m/s slows down at a constant rate of -3m/s/s
 - A) What is the velocity after 3 seconds of deceleration
 - B) What displacement does the car cover in this time: (Find X with $X_0 = 0$)

- 3) A runner moving with an initial velocity of 9m/s slows down at a constant rate of -1.5m/s/s over a period of 2 seconds.
 - A) What is her velocity at the end of this time
 - B) What displacement did she cover during this process(X with $X_0 = 0$)

- 4) if $V_0 = 3\text{m/s}$ and $V_{\text{final}} = 9\text{m/s}$ What is the average velocity if the acceleration is constant.

6) If the initial velocity of a car is 30m/s and its acceleration is 10m/s/s . Initial position is 0.

A) what its equation of motion $x=f(t)$.

B) what is the equation of $V=f(t)$)velocity as a function of time)

7) if the equation of motion of an object is $x = 5 + 10t + 2t^2$ What are:
Initial position ? initial velocity ? Acceleration ?

8) same question for $x = 10t + t^2$

9) $x = 10t^2$

10) if $x = 5 + 10t + 10t^2$

What is the velocity at $t=5s$ (after 5s)

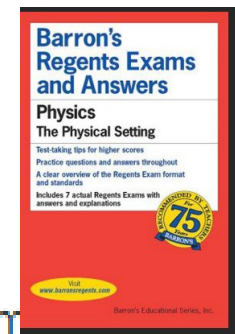
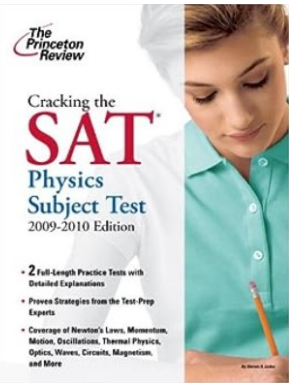
11) if $V = 10 + 2t$

What is the acceleration ?

What is the final velocity when $t=100s$ (so after 100s)?

What is the average speed during that time ?

- 1) How far does a car travel if it starts at rest and accelerated at 3m/s/s for **6.5s**
- 2) How long does it take a car, starting from rest, to travel 240m if its acceleration is 1.90m/s/s ? (in class)
- 3) How far a car travel if it starts at rest and accelerated at 3m/s/s until it reaches a speed of 22m/s ? (in class)
- 4) A car moving 22m/s has its brakes jammed on and leaves skid marks 45m long. What was its acceleration ?
(negative) (in class)
- 5) What is the acceleration of a rocket-driven sled that travels 360m in 5s , starting from rest and accelerating uniformly ?
- 6) How fast is a car going if it starts at rest and accelerates uniformly at 2.8m/s/s while traveling 220m ?
- 7) A car traveling in a straight line with an initial velocity of 14m/s accelerates at 2m/s/s to a velocity of 24m/s . Do in class
The final velocity is 24m/s ?
What is the time elapsed?



An object starts from rest at time $t = 0.00$ s and moves in the $+x$ direction with constant acceleration. The object travels 12.0 m from time $t = 1.00$ s to time $t = 2.00$ s. What is the acceleration of the object?

A jet plane is launched from a catapult on an aircraft carrier. In 2.0 s it reaches a speed of 42 m/s at the end of the catapult. Assuming the acceleration is constant, how far did it travel during those 2.0 s?

A car starts from rest and accelerates with a constant acceleration of 1.00 m/s^2 for 3.00 s. The car continues for 5.00 s at constant velocity. How far has the car traveled from its starting point?

The position $x(t)$ of a particle as a function of time t is given by the equation

$x(t) = (3.5 \text{ m/s})t - (5.0 \text{ m/s}^2)t^2$. What is the average velocity of the particle between $t = 0.30$ s and $t = 0.40$ s?

GRAPHICAL ANALYSIS OF MOTION

EQUATIONS OF KINEMATICS

for motion with velocity and acceleration = 0



1) $x = x_i + V_i t$ (or Δx)

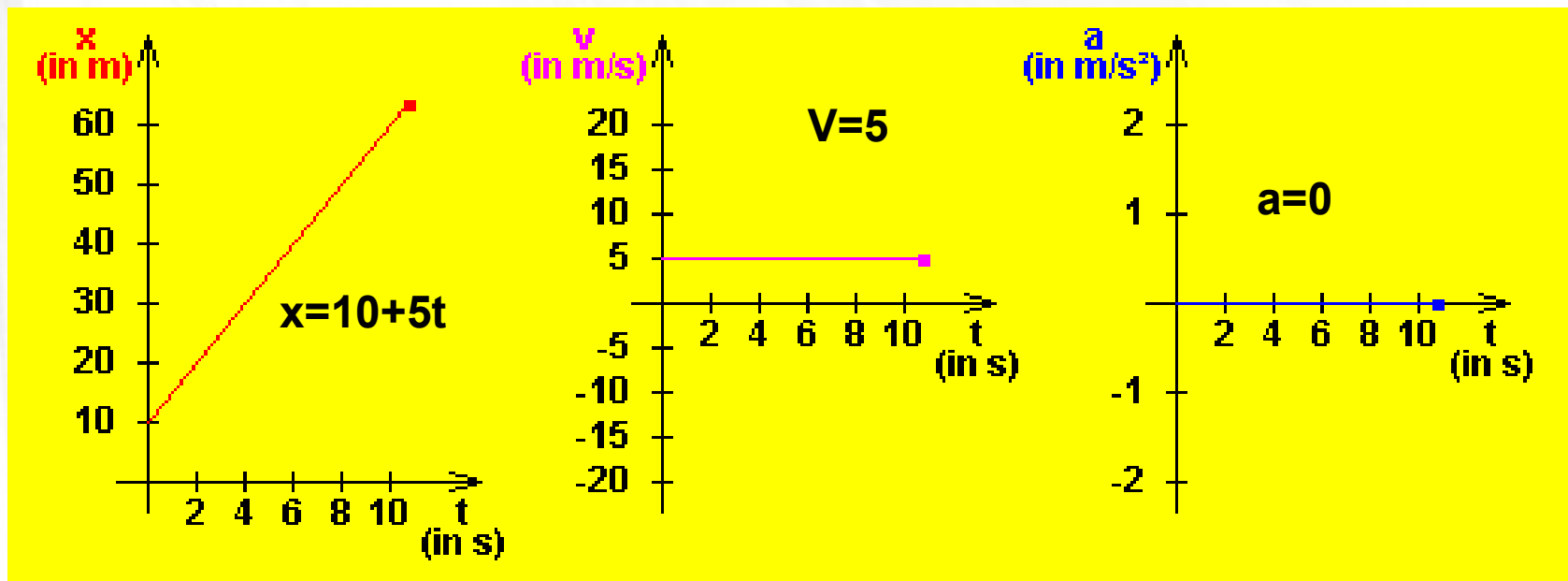
Position vs time $x(t)$ is a straight line. V_i is the slope. X_i the y-intercept

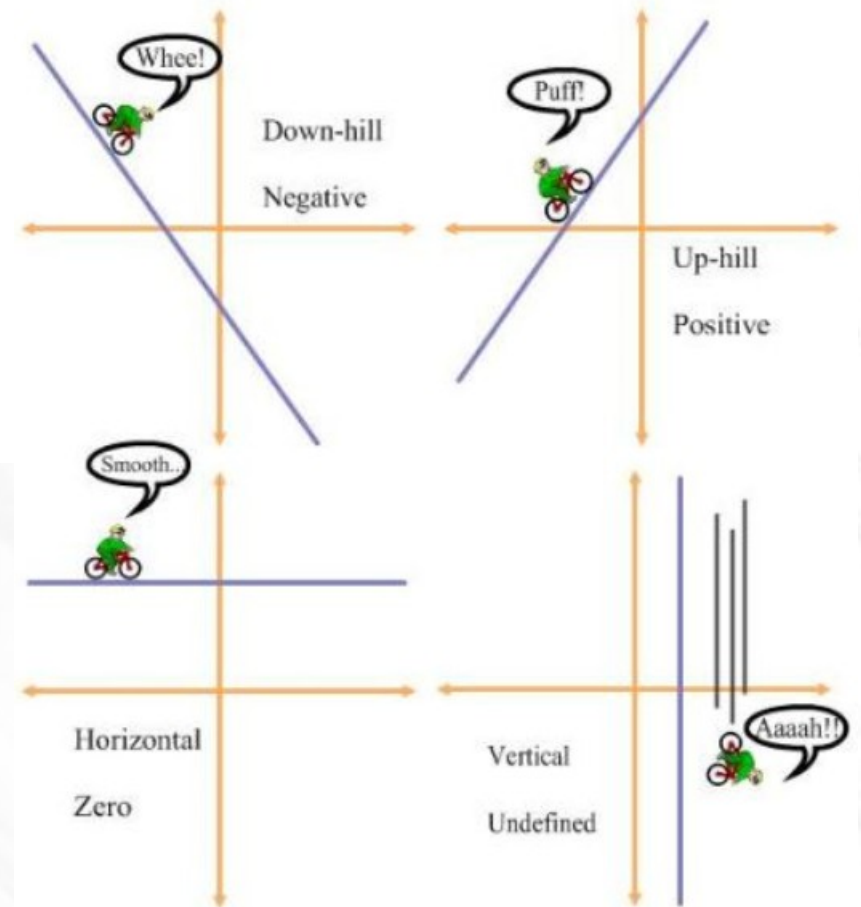
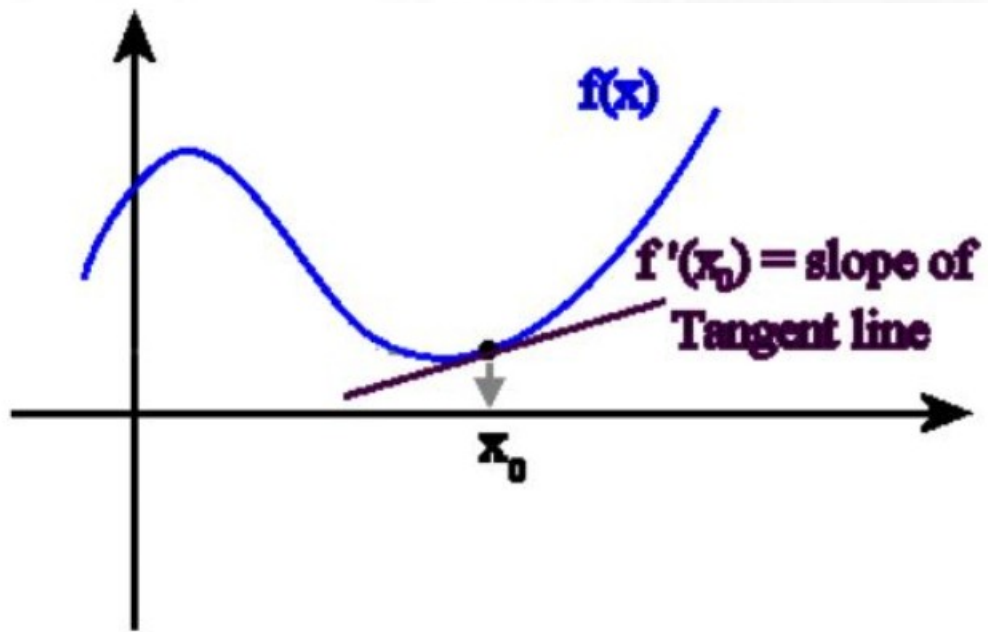
2) $V = V_i$ stays the same

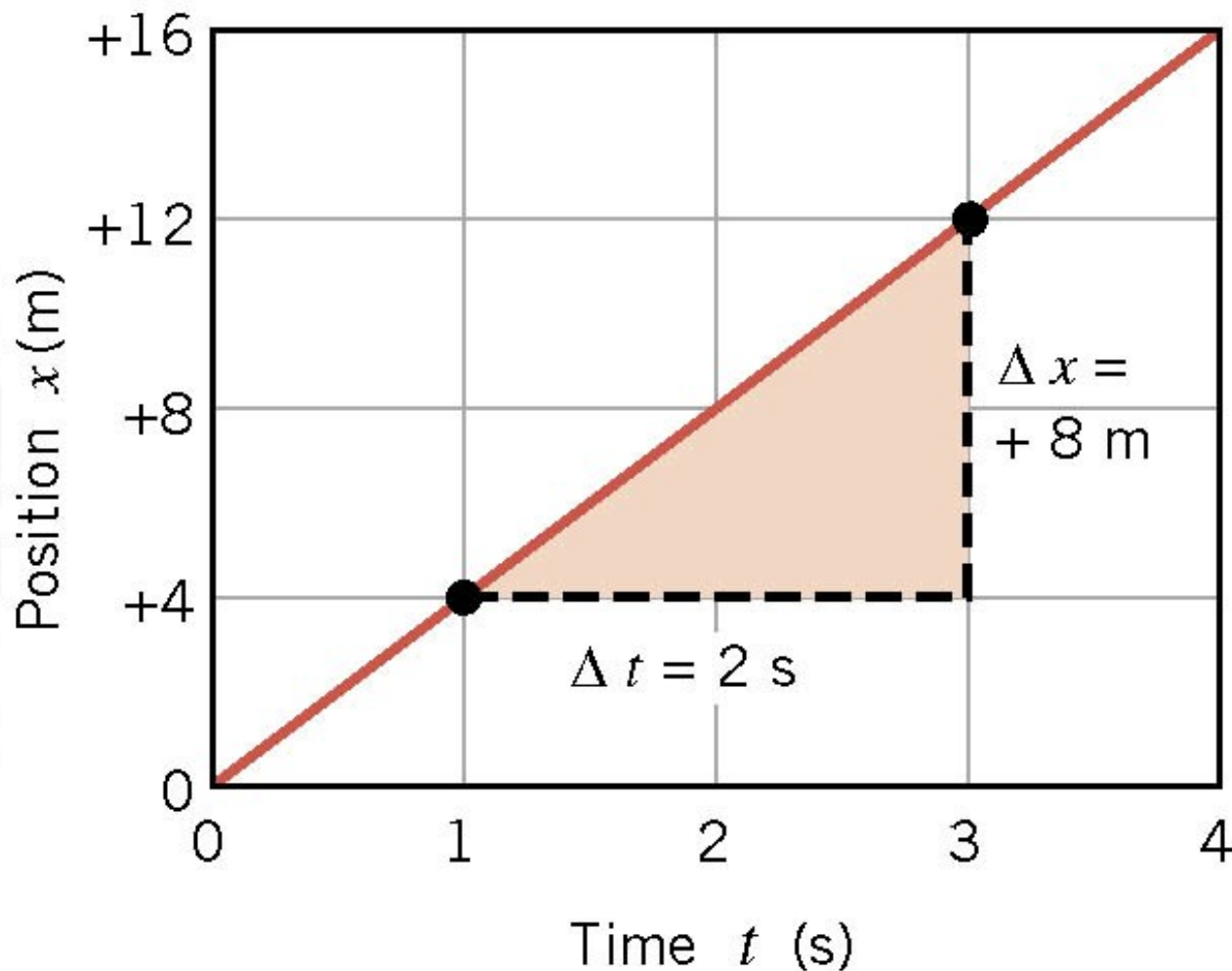
Velocity vs time $V(t) = V_0$ is a horizontal line

3) $a = 0$

Acceleration is 0







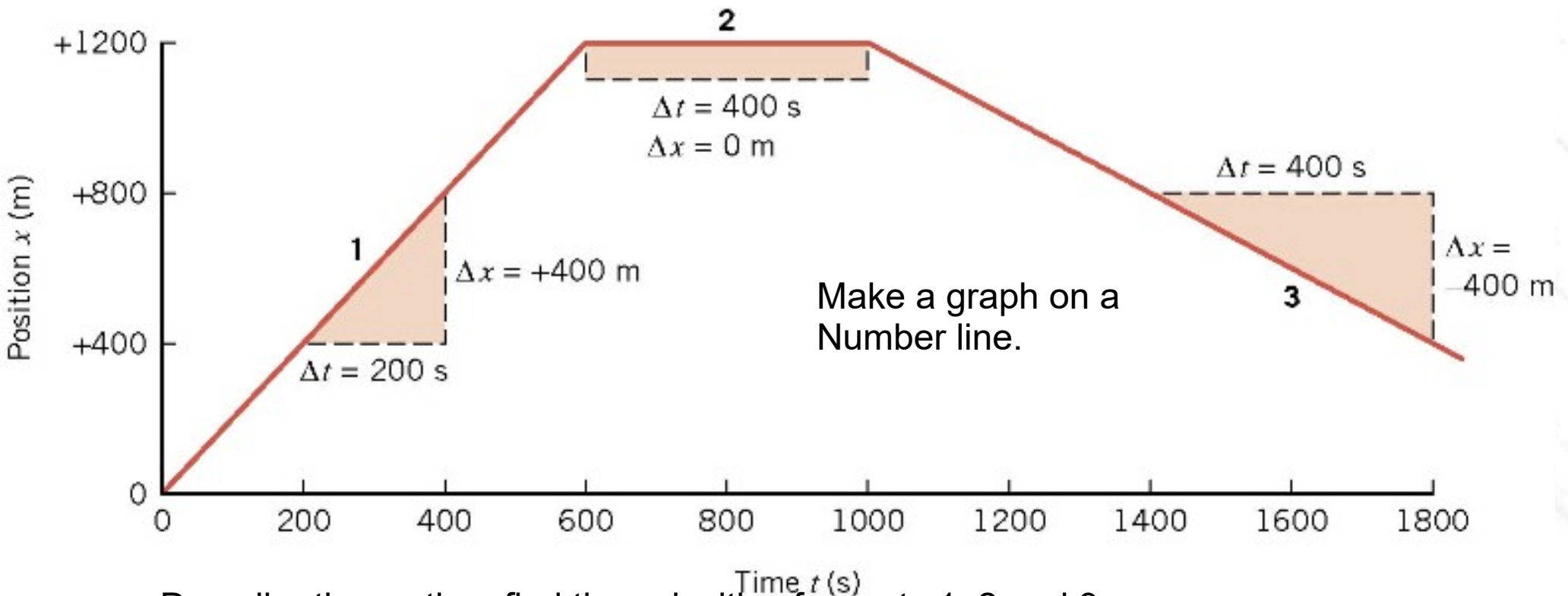
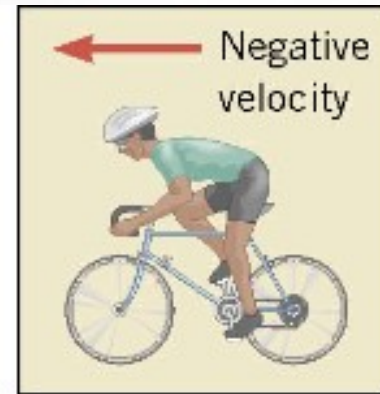
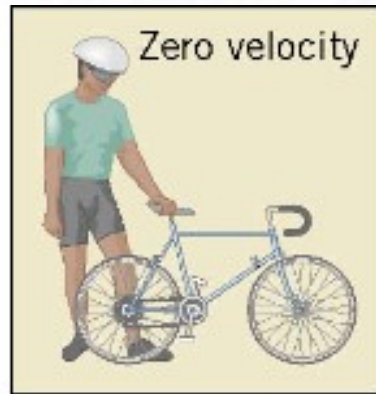
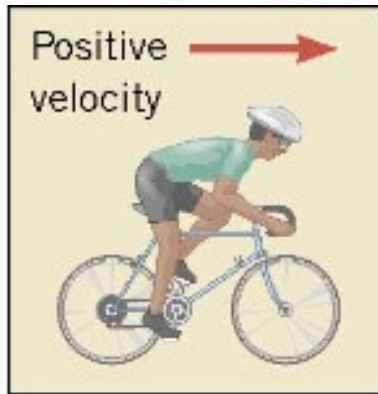
$$\text{Slope} = \frac{\Delta x}{\Delta t} = ?$$

What does the slope of the graph position vs time represent?

What is the unit ? What is the slope

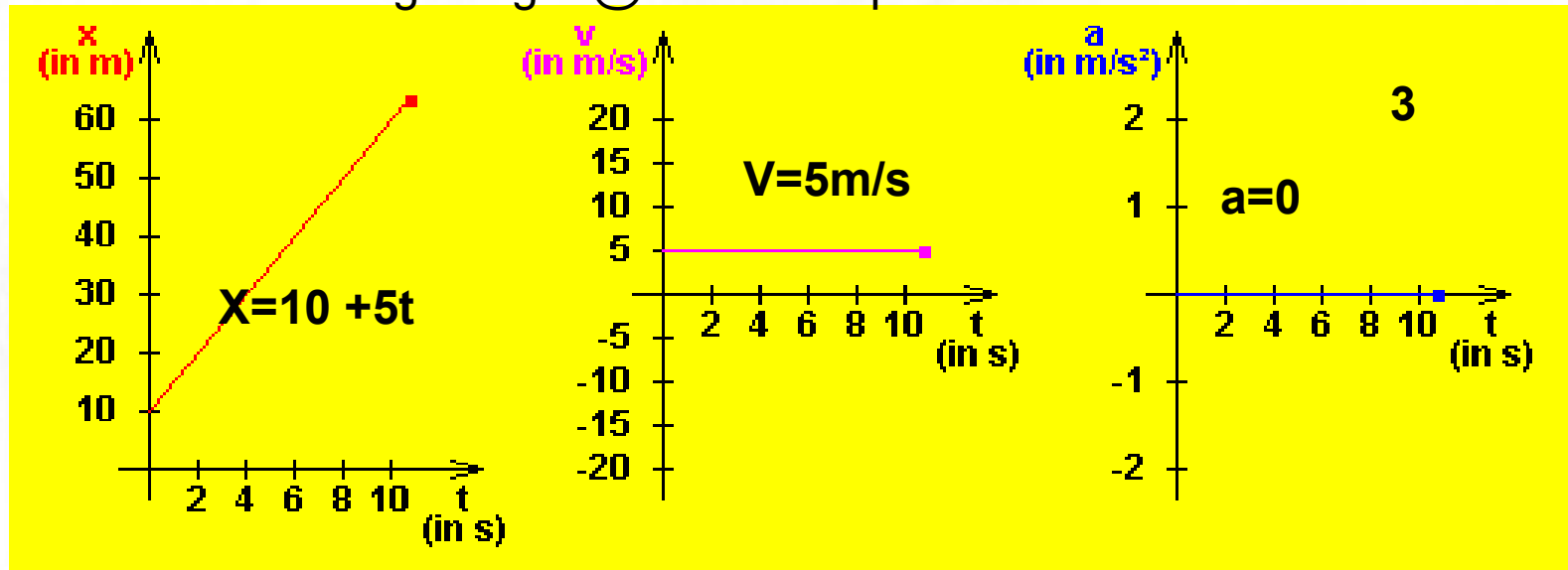
What is the the equation of motion $x(t)$

2.7 Graphical Analysis of Velocity and Acceleration

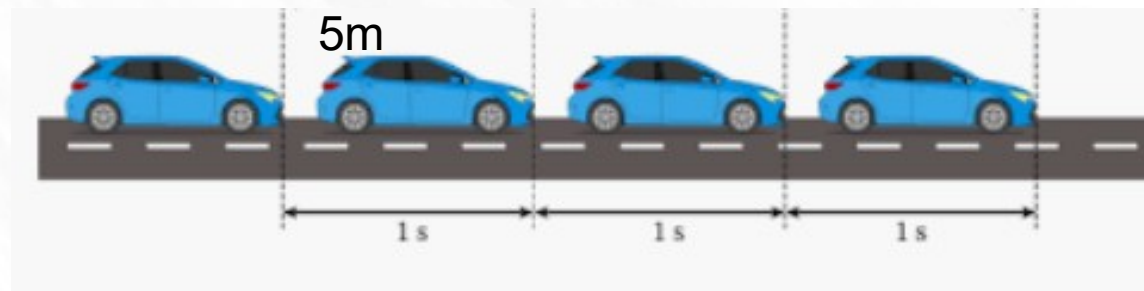


Describe the motion, find the velocities for parts 1, 2 and 3, don't forget the directions.

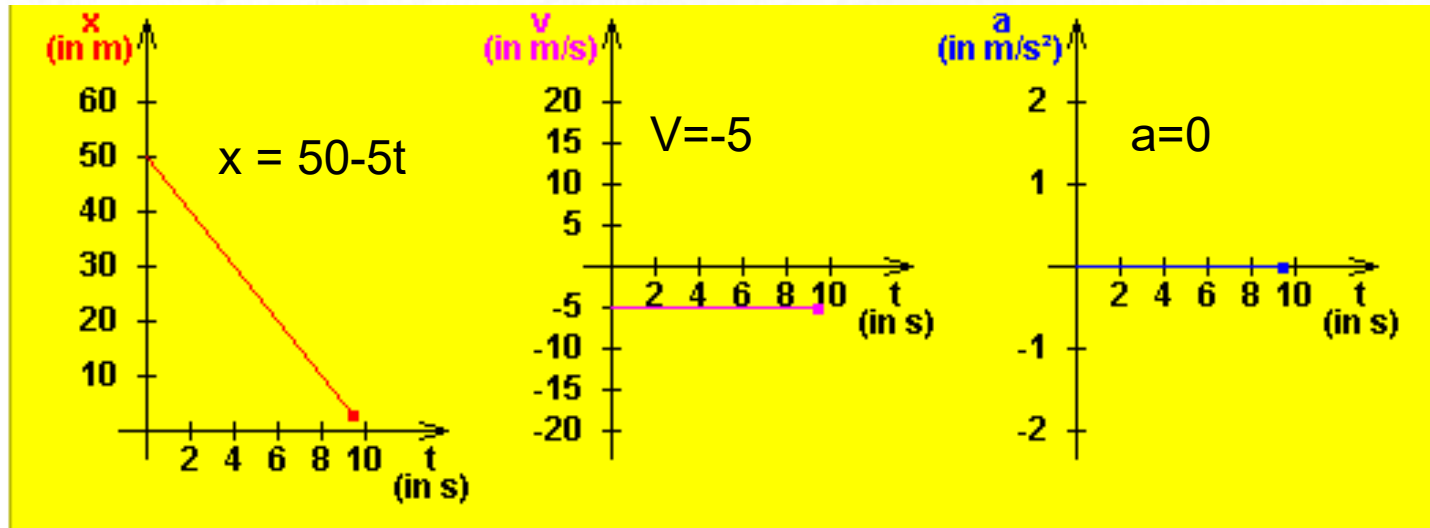
Moving at right @ constant speed



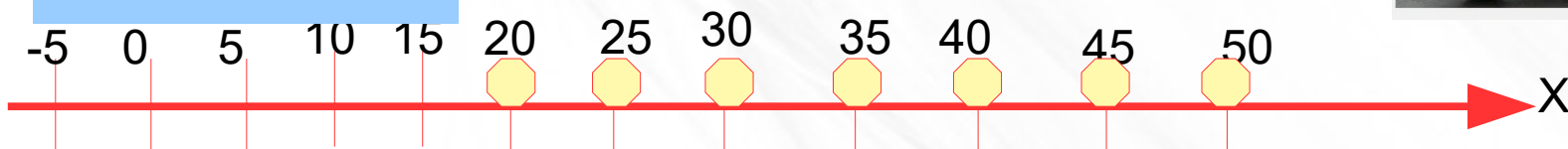
t(s)	x(m)
0	10
1	15
2	20
3	25
4	30



Moving @ left at constant speed



t(s)	x(m)
0	50
1	45
2	40
3	35
4	0



Complete the following statement: For an object moving at constant velocity, the distance traveled

- a) increases for each second that the object moves.
- b) is the same regardless of the time that the object moves.
- c) is the same for each second that the object moves.
- d) cannot be determined, even if the elapsed time is known.
- e) decreases for each second that the object moves.

EQUATIONS OF KINEMATICS for motion with constant acceleration

1) $x = x_0 + v_0 t + 0.5 a t^2$

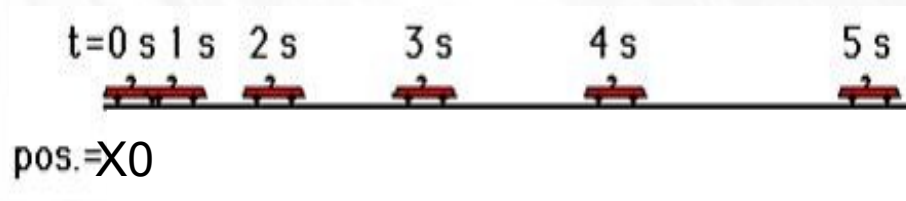
The graph $x(t)$ or position vs time is a parabola. The y-intercept is x_0 .
The slope (velocity) increases. The displacement increases every second.

2) $v = v_0 + a t$

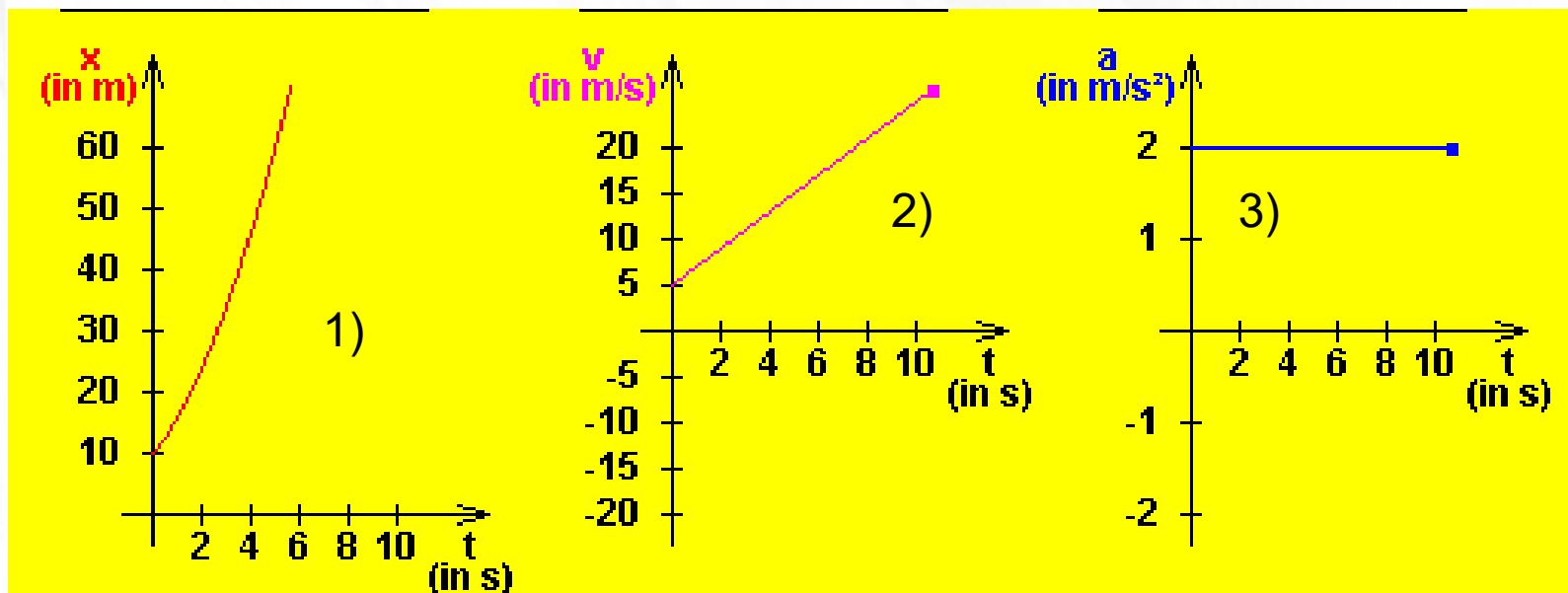
The graph is a line. The slope is the acceleration a . The y-intercept is v_0

3) $a = \text{constant}$

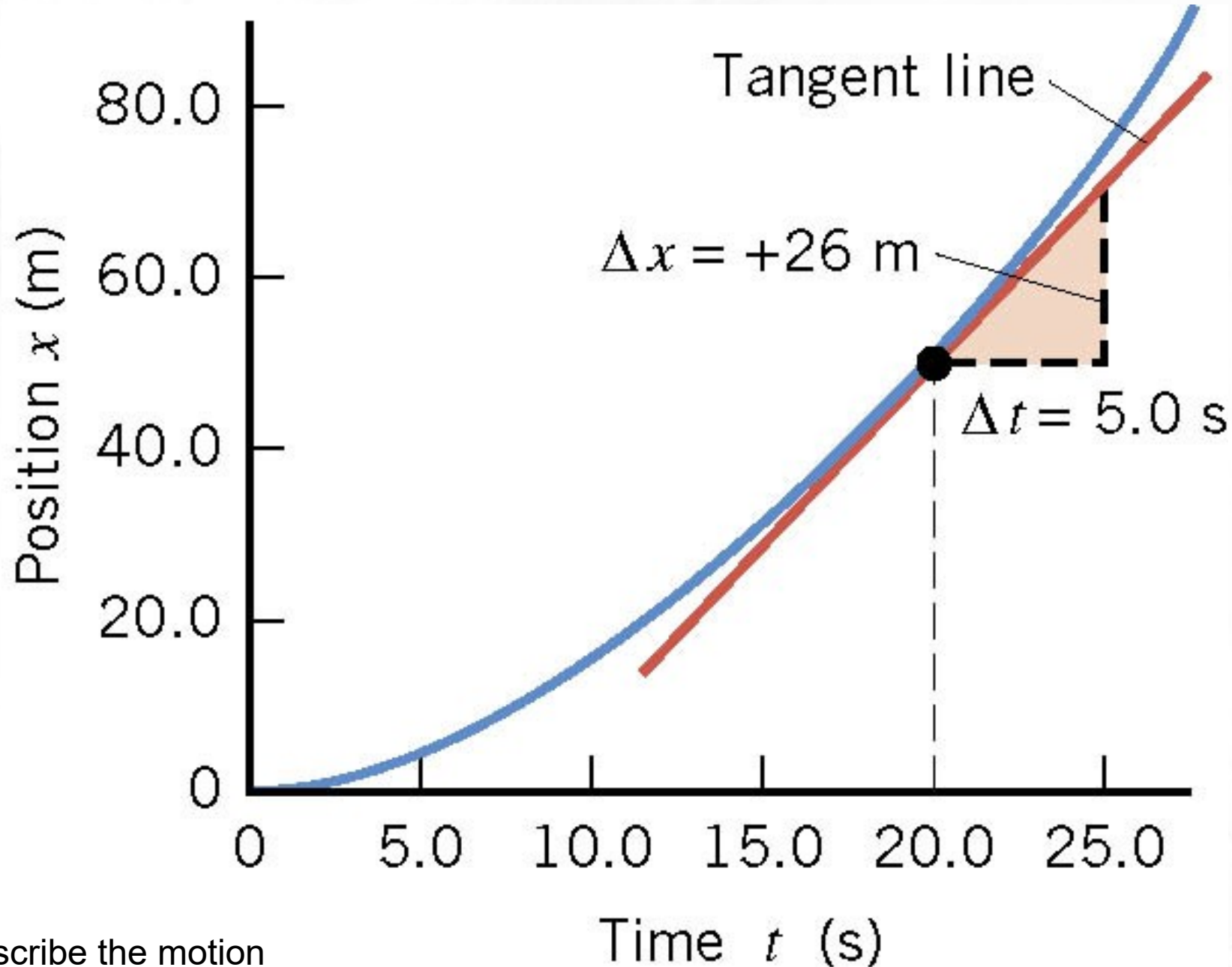
The graph is a horizontal line. a is constant.



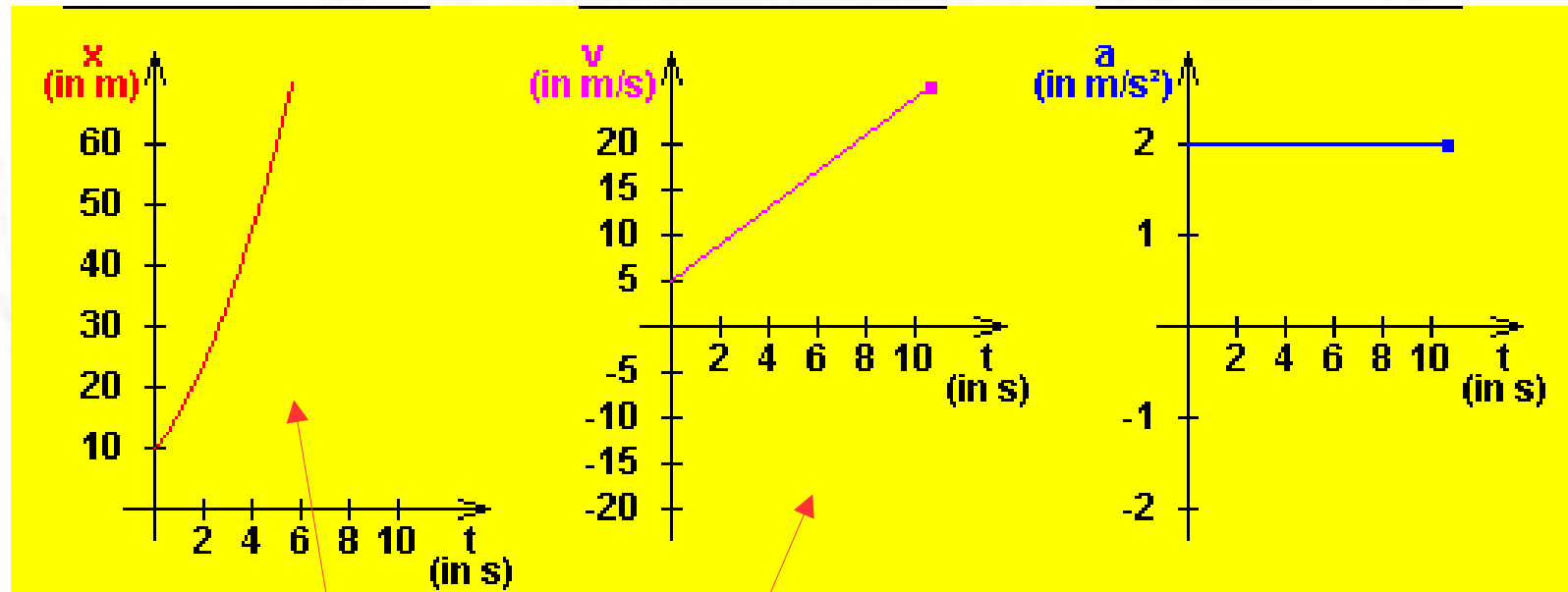
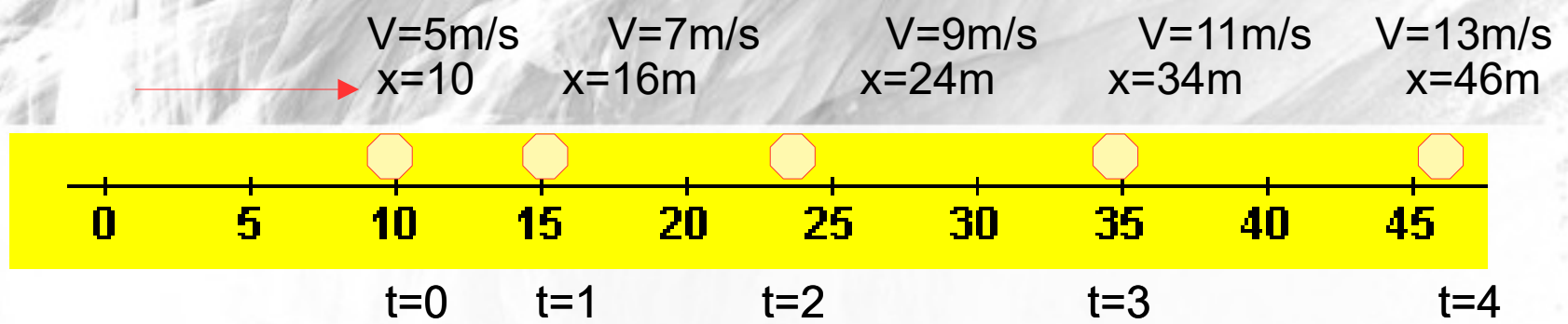
Show with Sketchbook the parabola



The slope of $x(t)$ is the velocity $V(t)$ at a given time t . It keeps increasing at a constant rate a .



Describe the motion
And find the instantaneous velocity



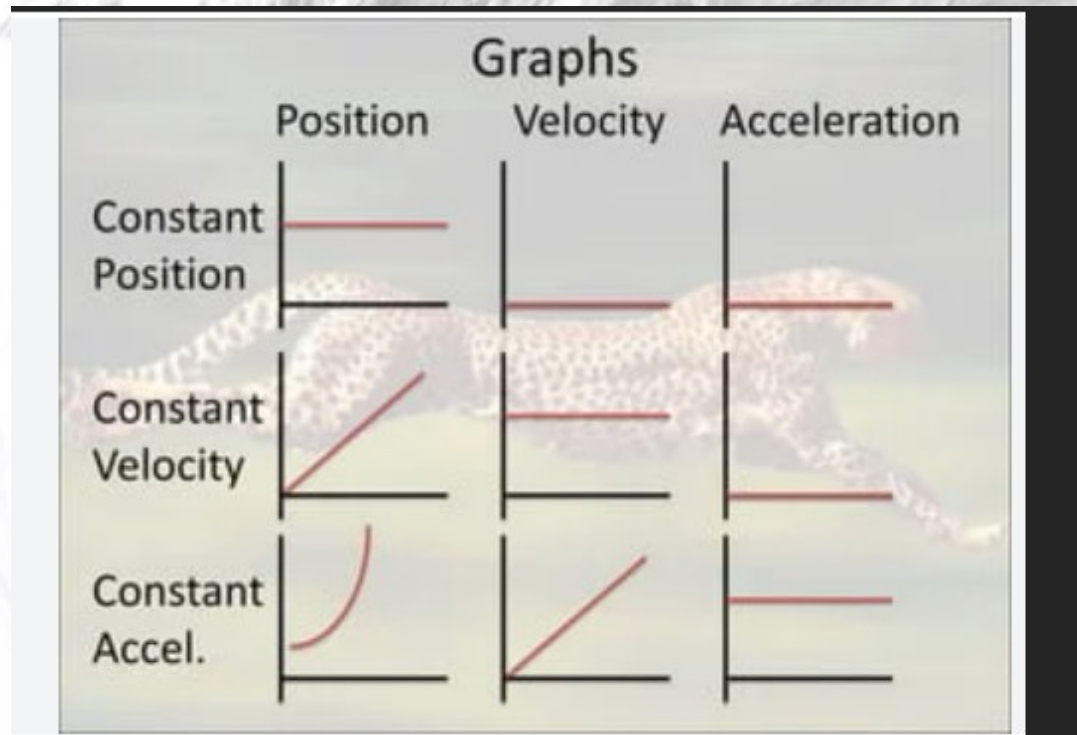
t(s)	x(m)
0	10
1	16
2	24
3	34
4	46

$$a=2\text{m/s/s}$$

$$V=5 + 2t$$

$$X = 10 + 5t + 0.5(2)t^2$$

t(s)	V(m/s)
0	5
1	7
2	9
3	11



Play the gif

slope $x(t)$ = velocity. If velocity > 0 its moving @ right (up).
 if velocity < 0 its moving @left (down). If slope $x(t)$ = constant $\rightarrow a=0$

If slope $x(t) = 0 \rightarrow$ not moving. If slope $x(t)$ changes $\rightarrow$ velocity is changing, there is an acceleration

slope $V(t)$ = acceleration.

If velocity and acceleration same direction $\rightarrow$ speeding up

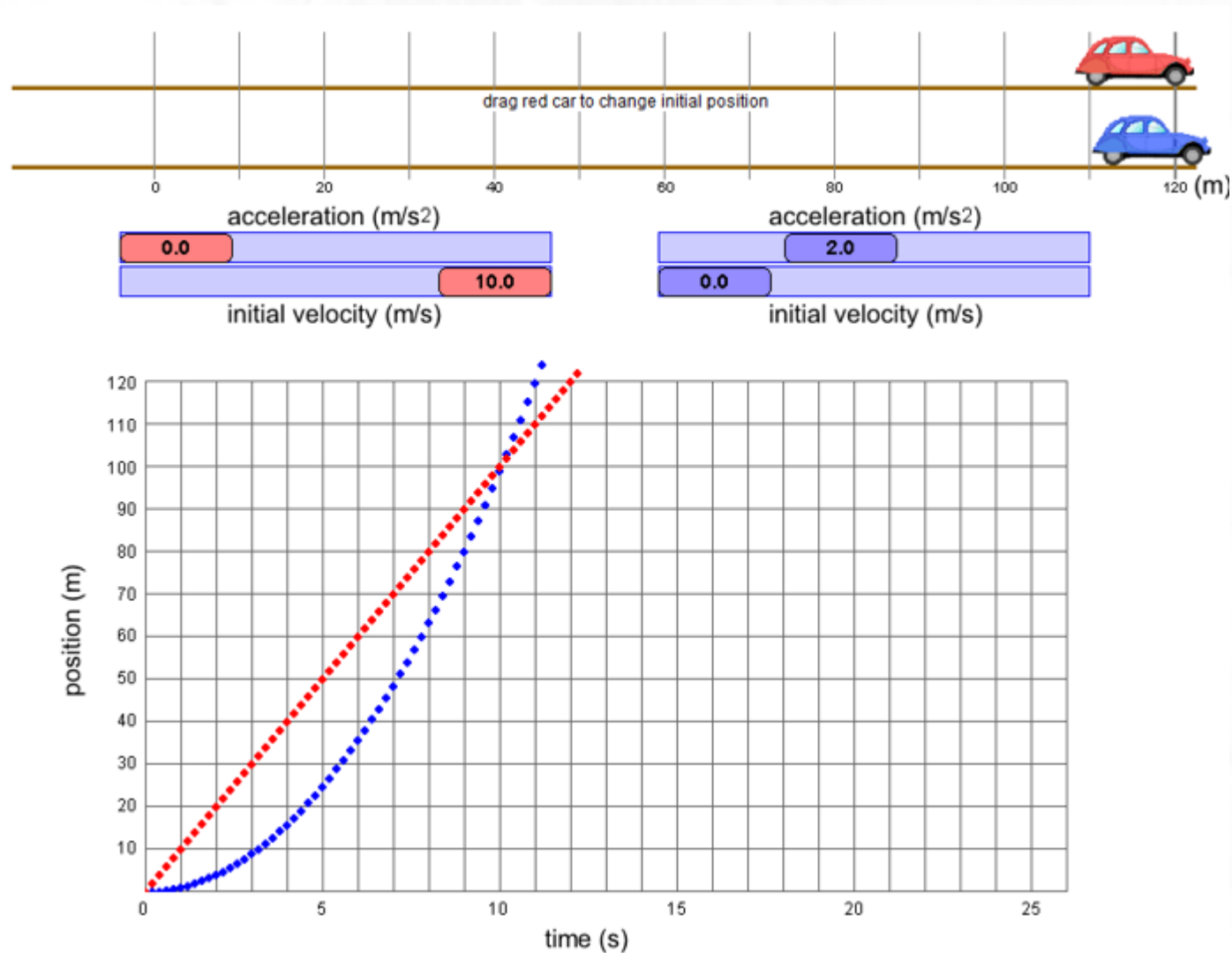
if velocity and acceleration not same direction $\rightarrow$ slowing down

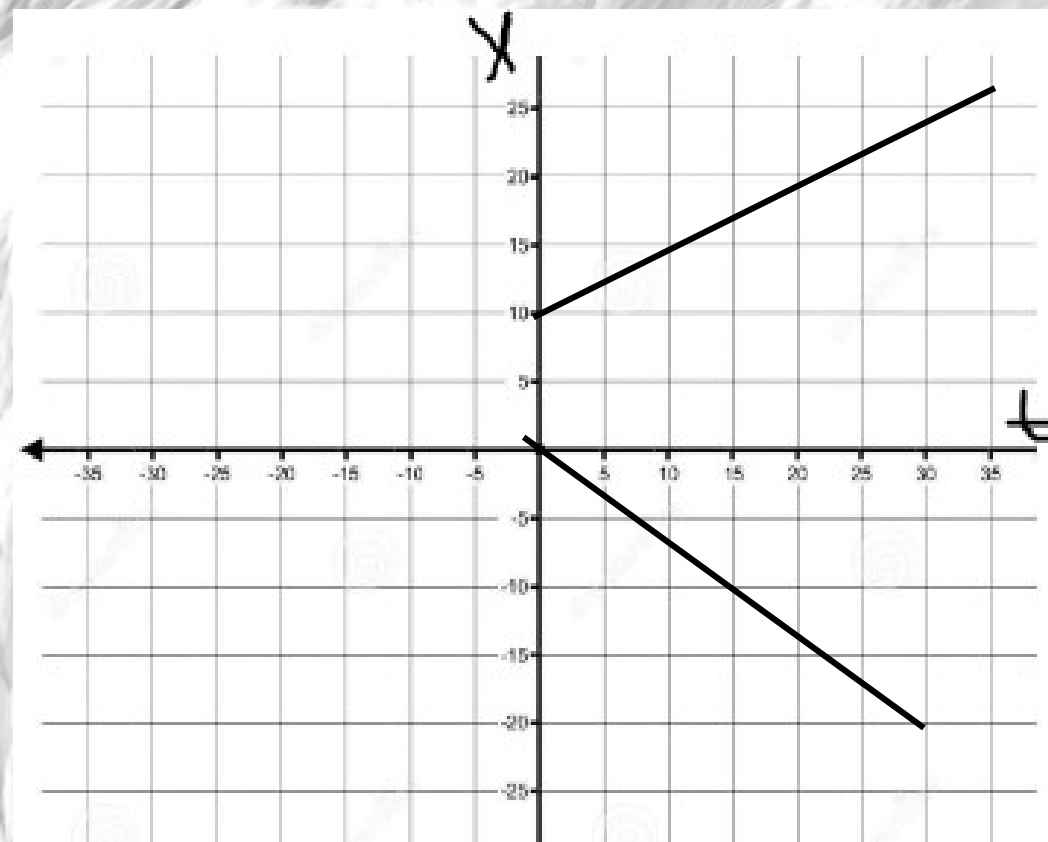
A slope is the steepness at a given time

See app.

How to find the time and place when they meet?

Find the kinematics equations





Equation of motion $X(t)$
And $V(t)$ for 2 graphs

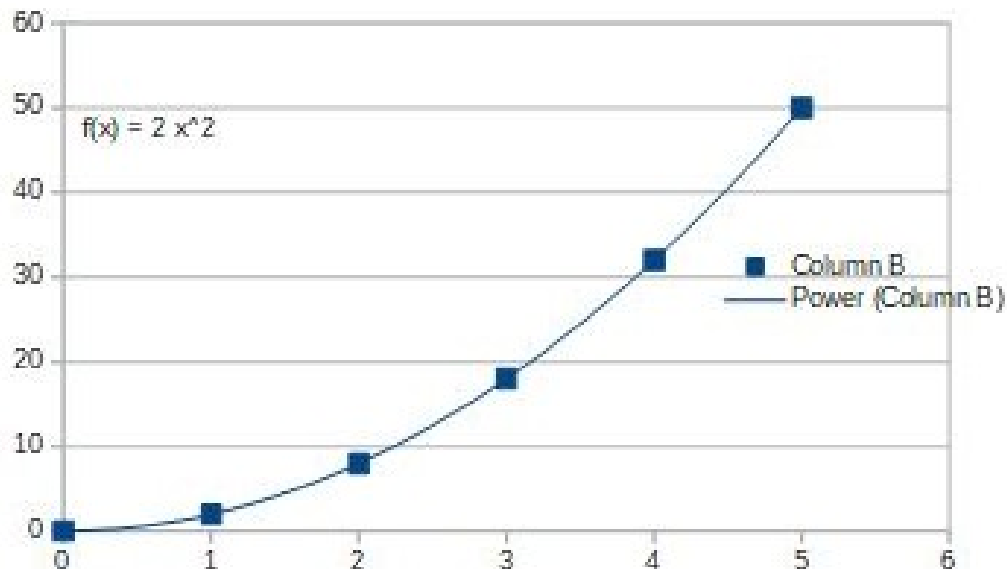
This is a graph position vs time
What is the equation of the motion
(use physical quantities x and t)

What is the average velocity between
 $t=2$ and $t=5s$? (use equ)

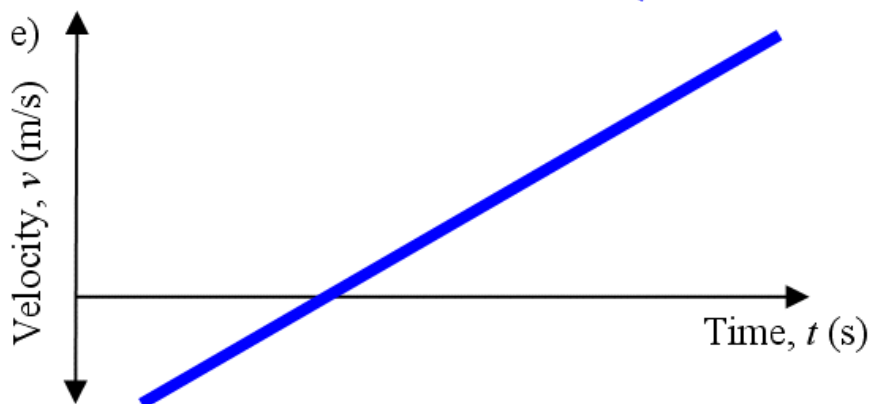
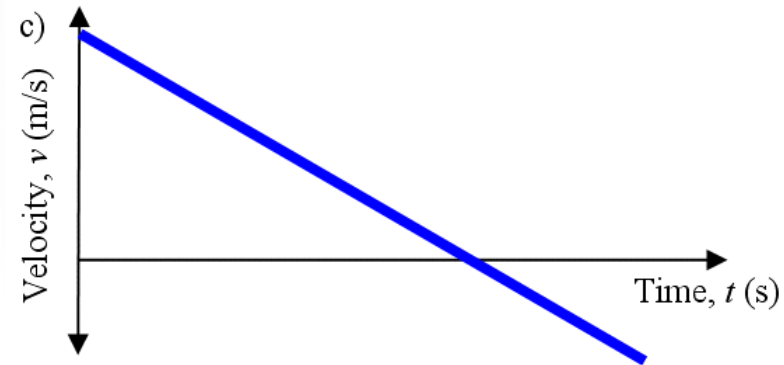
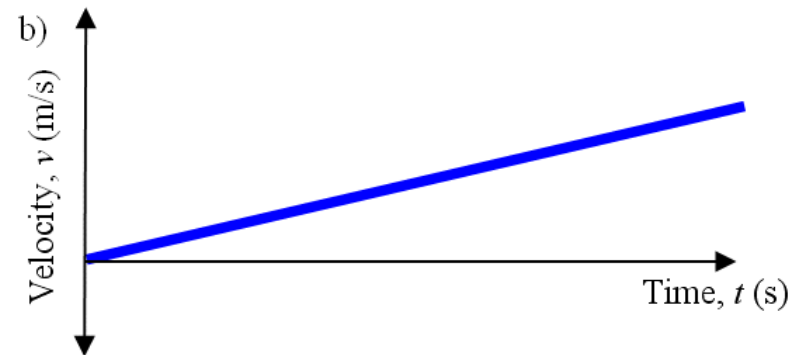
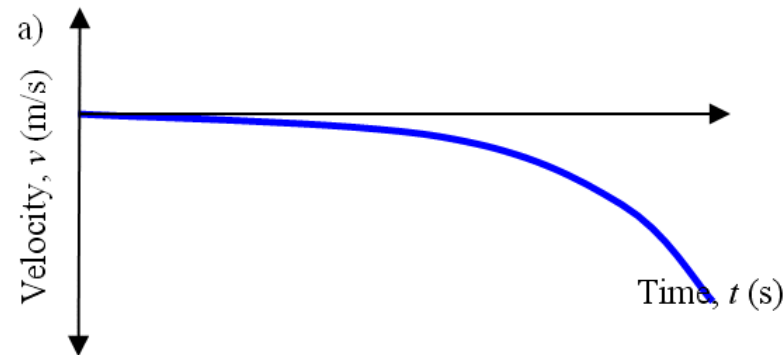
What is the position at $t=3s$? (equ)

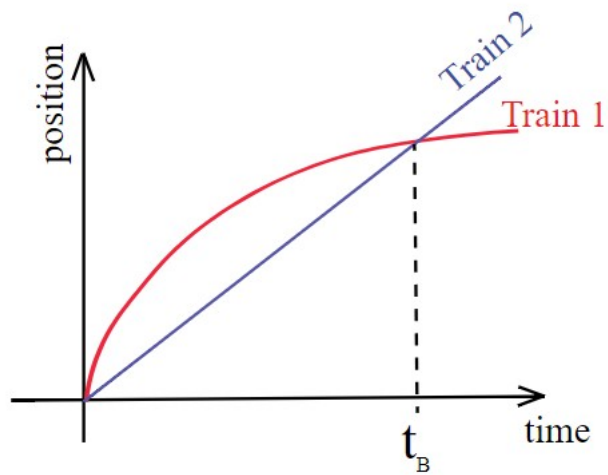
Velocity is the slope of the graph
What is V_0 ? (V at $t=0$ or slope
At $t=0$)

Use the equation to find the acceleration
And V_0



- . Which of the following velocity vs. time graphs represents an object with a negative constant acceleration?





Here are the positions of 2 trains.

1) describe the motion of train1 and train2
(constant vel ? Speeding up ? Slowing down ?)

2) What happens at t_B ?

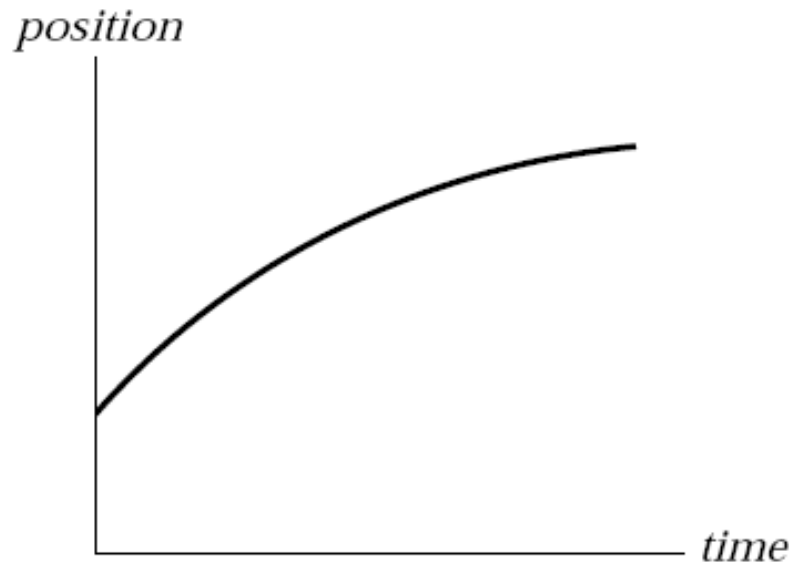
3) Can you see when the 2 trains have the same velocity ?

- . Which one of the following quantities can be determined from the slope of a position versus time graph for an object in motion?
- a) position
 - b) velocity
 - c) acceleration
 - d) distance traveled
 - e) displacement

Which one of the following quantities can be determined from the slope of a velocity versus time graph for an object in motion?

- a) position
- b) velocity
- c) acceleration
- d) distance traveled
- e) displacement

A train car moves along a long straight track. The graph shows the position as a function of time for this train. The graph shows that the train:



1. speeds up
2. slows down all the time.
3. speeds up part of the time and slows down part of the time.
4. moves at a constant velocity.

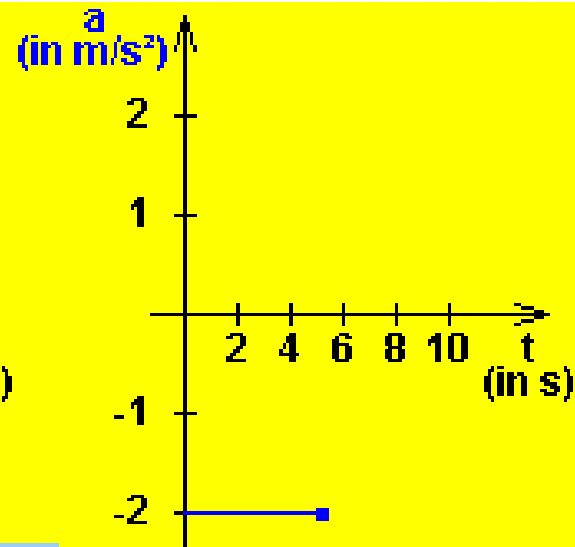
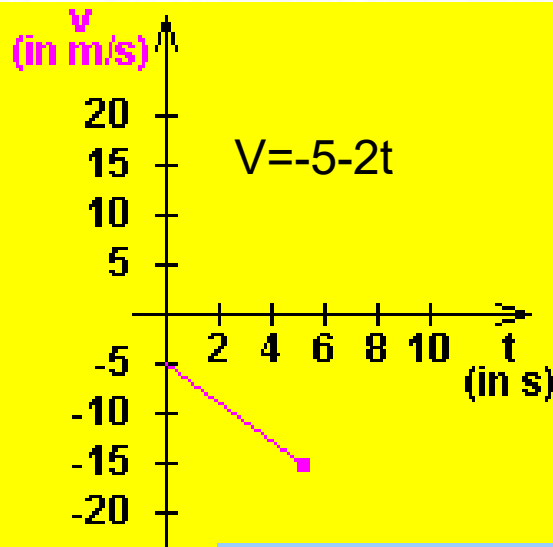
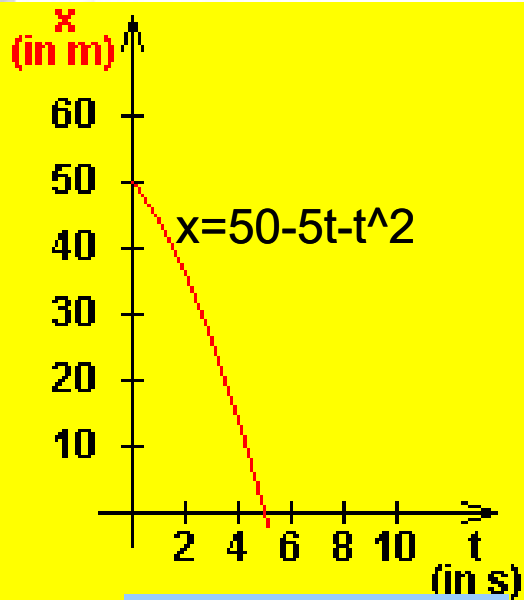
Speeding up @ down or speeding up @ left



$t=0$

Velocity and acceleration have the same direction → speeding up

Example: dropping a ball on a planet with gravity = -2m/s/s = acceleration.

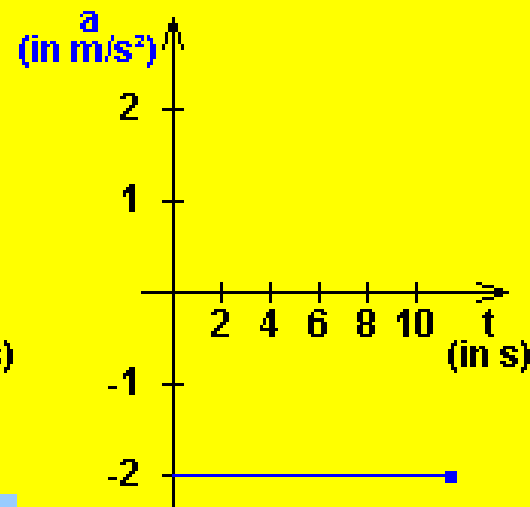
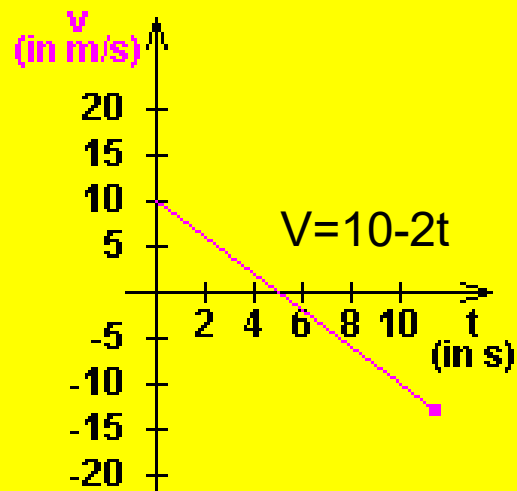
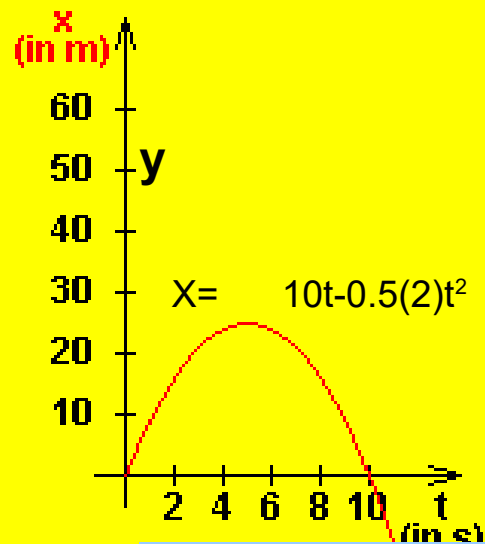


t	V
0	-5
1	-7
2	-9
3	-11

t	x
0	50
1	44
2	36
3	26



Consider the motion of an object thrown in the air with an initial velocity V_0 . It moves @ up but slows down, stops and changes direction and moves @ down But accelerates. $y(t)$ is the position of the object. Its on another planet

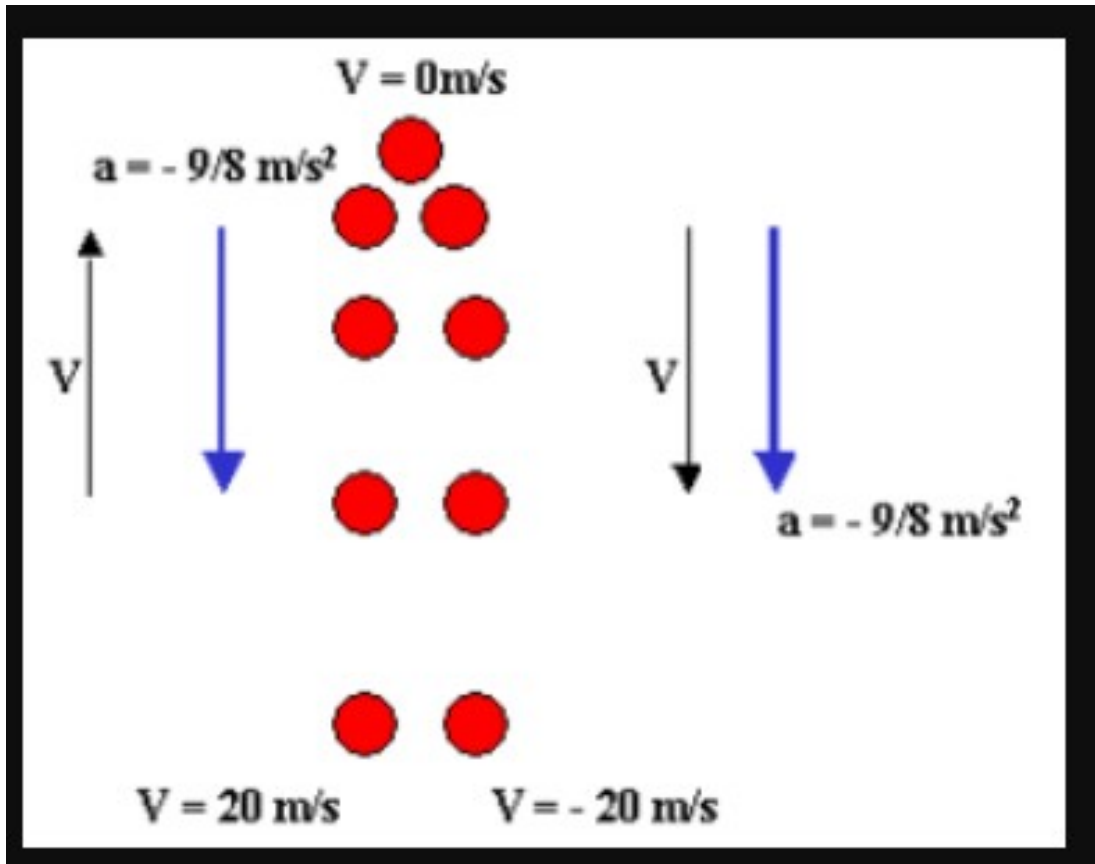


t	x
0	0
1	9
2	16
3	21
4	24
5	25
6	24
7	21
8	16
9	9
10	0

t	V
0	10
1	8
2	6
3	4
4	2
5	0
6	-2
7	-4
8	-6
9	-8
10	-10



Similar situation:



Throwing a ball straight up.

The weight (force) is always pulling @ down. So acceleration is always @ down (negative)

On the way up the force causes The ball to slow down.

On the way down the force causes The ball to speed up.

Find the kinematics equations.